

# Book Cover





DOINGG@UNION.EDU

COURSE: CSC:483 SPECIAL TOPICS - AP-  
PLIED BIOLOGICAL DATA SCIENCE  
DOCUMENTATION

SELF

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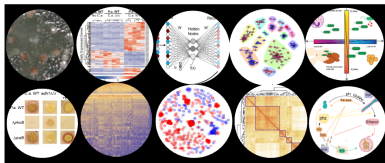
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# Home



The course website for [CSC483](#), part of the Union College [CS curriculum](#). Here you can find our weekly schedule, assignments and resources.

## Announcements

Check-out HW9 to schedule your final presentation!

## Office Hours

- Student/Office Hours:
  - Steinmetz 108B
  - Wednesday 2:00 pm - 3:30 pm
  - Thursday 4:00 pm - 5:30 pm
  - subject to change, check course website for most up-to-date schedule
  - drop-in or schedule a 15 minute slot: <https://calendar.app.google/8bus6pfDvyphR9ar5>
  - by appointment for another time or over zoom

*Course Description*

In this course we will gain familiarity and practice in methods of analyzing biological sequence data and the large datasets that have resulted from decades of high-throughput experimentation. We will follow how data are turned into biological knowledge through the general workflow of (1) sequence alignment, (2) dimensionality reduction and (3) statistical, comparative hypothesis testing. At each stage we will compare state-of-the-art deterministic methods with emerging machine learning-based counterparts and interrogate trade-offs in performance and interpretability. We will explore the role of interdisciplinary communication in biological data science and the iterative process of program development and implementation in exploring the great biological unknowns. This course will use Python and R and introduce libraries useful in applied computer science and guidance on how different approaches are used in different contexts. No prior biological knowledge will be assumed.

Prerequisite(s): MTH 197 and a C- or higher in CSC 151. Recommended: CSC 250. MTH 199 can be substituted for MTH 197 CC: SET Note: Course can be repeated for credit under different topics. Consult with the department chair for more information.

# Schedule

Weekly Schedule

## Schedule

| W  | TOPIC                                  | Due                             | Notes                   |
|----|----------------------------------------|---------------------------------|-------------------------|
| 1  | Scope of Biological Data               | HW 01                           |                         |
| 2  | Arc of Analyses                        | HW 02                           |                         |
| 3  | Sequence Alignment                     | HW 03                           |                         |
| 4  | Data Wrangling                         | HW 04                           | Exploratory<br>Analyses |
| 5  | Dimensionality Reduction               | Exploratory<br>Plotting         |                         |
| 6  | Visualization                          | Project Proposal                |                         |
| 7  | Hypothesis Testing                     | Results/Figures                 | Drafts                  |
| 8  | Peer Review                            | Peer Review                     |                         |
| 9  | Methodological Comparative<br>Analysis | Response to<br>Reviewers        |                         |
| 10 | Project Presentations                  | Final Project &<br>Presentation |                         |



## *Resources*

### *Rosalind Probblems*

- Course enrollment link: <https://rosalind.info/classes/enroll/750c73565c/>

### *Google Colab Pro*

- Pro account sign-up: <https://colab.research.google.com/signup>
- Use your Union email

### *Late Token Form*

- These are for project-related deadlines, not Rosalind problems
- These can be used for any reason
- These do not apply to your final deadline
- Form link: <https://forms.gle/ZcEss9oNayecL8zr7>

### *Python Style Guide*

- PEP 8
- should be used in all code cells of Colab notebooks
- Link: <https://peps.python.org/pep-0008/>



*Syllabus*

*Syllabus*



# CSC483-Special Topics - Biological Data Science Syllabus

## 1 CSC 483: Special Topics - Applied Biological Data Science

- Union College, Winter 2026
- Georgia Doing, PhD
- email: doingg@union.edu
- office: Steinmetz 108B

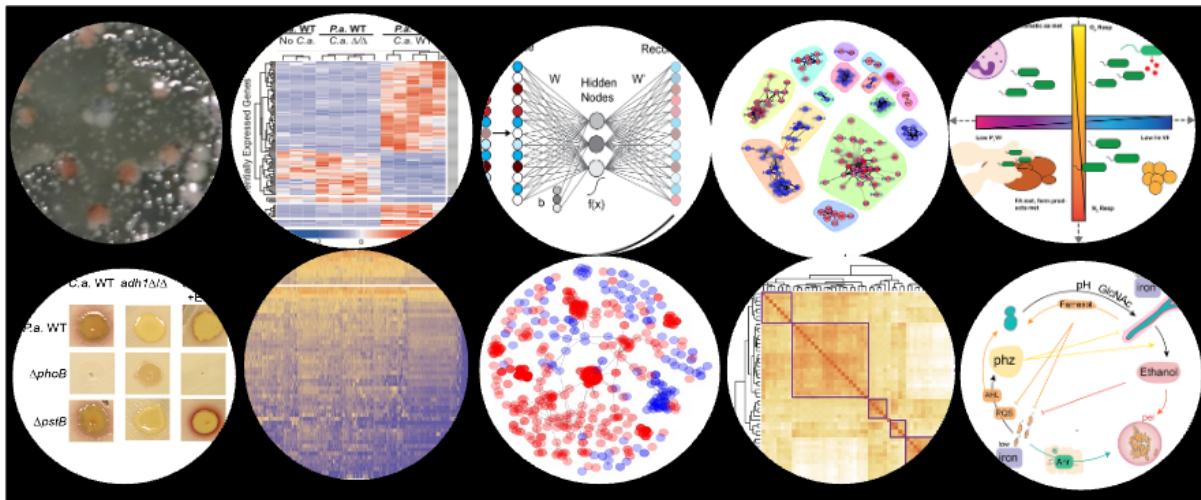


Figure 1: Views of Biological Data.

## 2 COURSE BASICS

- Lecture: Tuesday/Thursday 1:55 am - 3:40 pm
- Location: ISEC 070
- Student/Office Hours:
  - Wednesday 2:00 pm - 3:30 pm
  - Thursday 4:00 pm - 5:30 pm
  - subject to change, check course website for most up-to-date schedule
  - drop-in or schedule a 15 minute slot: <https://calendar.app.google/8bus6pfDvyphR9ar5>
  - by appointment for another time or over zoom

## 3 OVERVIEW

In this course we will gain familiarity and practice in methods of analyzing biological sequence data and the large datasets that have resulted from decades of high-throughput experimentation. We will follow how data are turned into biological knowledge through the general workflow of (1) sequence alignment, (2) dimensionality reduction and (3) statistical, comparative hypothesis testing. At each stage we will compare state-of-the-art deterministic methods with emerging machine learning-based counterparts and interrogate trade-offs in performance and interpretability. We will explore the role of interdisciplinary communication in biological data science and the iterative process of program development and implementation in exploring the great biological unknowns. This course will use Python and R and introduce libraries useful in applied computer science and guidance on how different approaches are used in different contexts. No prior biological knowledge will be assumed.

Prerequisite(s): MTH 197 and a C- or higher in CSC 151. Recommended: CSC 250. MTH 199 can be substituted for MTH 197 CC: SET Note: Course can be repeated for credit under different topics. Consult with the department chair for more information.

## 4 LEARNING OBJECTIVES

By the end of the course, you should be able to *answer* the following questions:}

- How is sequence data transformed into quantitative data?
- What are the fundamentals of data wrangling?
- What are best practices in data visualization?
- How can data be communicated between data scientists and biologists?

By the end of the course, you should be able to *do* the following:

- Acquire and prepare data for analysis.
- Conduct descriptive data exploration.
- Communicate findings of an independent project.

A 1-page schedule-at-a-glance is included at the end of the syllabus, but please see the course website more detailed schedule.

## 5 CLASS POLICIES AND GUIDELINES

The Computer Science Department as a whole welcomes all people, regardless of age, background, beliefs, ethnicity, gender identity, gender expression, national origin, religious affiliation, sexual orientation and any other differences, be they visible or non-visible. It's also important to recognize that institutional racism has prevented members of marginalized groups - especially black, indigenous and other people of color (BIPOC) - from fully participating in the field of Computer Science.

***You. Yes you, the one reading this syllabus. You belong here.***

As an instructor, I will do my utmost to uphold these principles and to treat everyone with respect. As students in this class, I expect you to do the same since one person is not a community unto themselves. We're all in this together, so let's treat each other well. Lastly, I welcome feedback on issues we might discuss or ways that our class can be a more just one for all people.

## 6 COURSE MATERIALS

Online Textbook: Runestone (free).

We are going to use an online, interactive, free and open source textbook. This textbook has a lot of embedded interactive exercises. Reading this book always also means doing the activities. To get access to the book, please register on Runestone using the following information which can also be found on Nexus:

- <https://runestone.academy/runestone/default/user/register>
- your Union College email
- unioncollege\_py4e-int\_fall25 as the course name

We'll use the following websites and software throughout the course:

- Rosalind problems: <https://rosalind.info/classes/enroll/750c73565c/>

- The course website: <https://georgiadoing.github.io/tufte-quarto/>
- Google Colab: the cloud-based platform to write and run Jupyter Notebooks with python code
  - Pro account sign-up: <https://colab.research.google.com/signup>
  - Use your Union email

## 7 COURSE RESOURCES

The course website has a list and links to the readings and software resources that we are going to use. In class, we are going to use the Linux machines in ISEC 070. Outside of class you have a choice of hardware. You can access Google Colab via a browser on your own machine, install the software on your own computer (Python is freely available for Windows, Mac, and Linux) or you can work in one of the Computer Science labs. We have three spaces that you can use 24/7 using your ID card, except when classes are being held in them:

- Olin 107
- PASTA Lab (ISEC 051+)
- Computer Science Resource room (Steinmetz Hall, 209A)

## 8 LIBRARY RESOURCES

The library is available to help you with your research needs! Librarians can help you develop research questions, search for and select the best sources for your projects, identify research strategies, evaluate sources, and assist you with creating citations. There are multiple ways for you to contact a librarian. For more information, please see our Ask A Librarian page: <https://www.union.edu/schaffer-library/ask-a-librarian>.

## 9 ACCOMODATIONS

It is the policy of Union College to make reasonable accommodations for qualified individuals with disabilities. If you are a person with a disability and wish to request accommodations to complete your course requirements, please make an appointment with me or stop by during my office hours as soon as possible, all discussions will remain confidential. You must provide reasonable notice and be in touch with Accomodative Services on the 2nd floor of Schaffer Library, if you have not already, if you wish to take advantage of extra time on exams.

## 10 GRADING

The following components will contribute to your final grade for this class with roughly the indicated weight.

- Final project: 25%
- Final oral exam: 25%
- Problem sets: 25%
- Peer review: 25%

Letter Grade Scale:

- A: 93-100; A-: 90-92
- B+: 87-89; B: 83-86; B-: 80-82
- C+: 77-79; C: 73-76; C-: 70-72
- D: 60-69; F: 0-59

## 11 LATE POLICIES

Homework assignments will usually be discussed in class on the day that they are due and thus, without prior agreement, there is NO late submission of assignments allowed. You have three “extension tokens” that can be used for due dates related to your final project for this class. Each token will extend a single project deadline by 24 hours. You can use each token on a different deadline, or use two or even all three on a single deadline. Once all three tokens have been used, no late submissions will be accepted (barring exceptional circumstances).

<https://forms.gle/ZcEss9oNayecL8zr7>

## 12 ATTENDANCE

Class participation is a critical component of the course and attendance is mandatory. I realize that sometimes other things come up (interviews, athletic events, illness, etc.). In those cases, just let me know (in advance, if at all possible) that you are going to be absent. You may not receive credit for make-up material if you did not discuss your absence with me prior to class. No matter how much class you have missed, please do not come to class if you have an illness that is likely contagious, please email me to work out a reasonable solution.

If you do miss class, it is your responsibility to make up any material that you missed. Get notes from a classmate, make sure to complete all homework assignments due or assigned

during the class that you missed, and come see me if you have any questions on the material. Unless you have made an arrangement with me ahead of time or you had an emergency, I will expect you to hand in all assignments by the due date, regardless of whether you were in class. You will not be able to make up missed exams or in-class questions and assignments.

Attendance in class constitutes being present, respectful and engaged. Accessing off-topic websites or software, checking email or phones, *utilizing large language models* or otherwise attending to things not pertinent to the course is distracting to yourself and others and will result in a reduction of your overall grade, up to a 0.5% reduction per class.

## **13 PARTICIPATION AND ENGAGEMENT**

Following these guidelines will constitute positive engagement:

- Respect. This course is a space for rigorous and respectful debate. When confronting ideas and people different from you, lead with curiosity rather than judgement. This commitment extends to all our face to face and digital interactions. We will critique ideas, not people. To protect our shared learning environment, you may not record, photograph, or share any part of our class sessions outside of our class.
- Show up to class on time and prepared. Show up ready to learn by completing the readings and exercises. When you are late or unprepared, you are disrespecting the learning experience for the group.
- Embrace intellectual humility. Recognize that there are limitations to your knowledge and that some of your beliefs could actually be wrong. Be curious about your thinking and open to learning from others. This is hard, but so vital to your learning in this course—we'll work on developing this throughout the term!
- Get help when you need it. If you are stuck or confused or lost, be proactive and get help. The best learners and thinkers can figure out when they are stuck and make a plan to get unstuck. Come to Office Hours (Wed/Thurs 2:00-3:30pm Steinmetz 108B), the CS Helpdesk (Sun-Thurs 7:00-9:00pm Olin 107) or ask another student. Often the best person to explain something is the person who just figured it out. Try reaching out to another student in the course who seems like they get it—they will probably be flattered you asked!

## **14 ACADEMIC INTEGRITY AND "ARTIFICIAL INTELLIGENCE" USE**

Union College recognizes the need to create an environment of mutual trust as part of its educational mission. Responsible participation in an academic community requires respect for and acknowledgement of the thoughts and work of others, whether expressed in the

present or in some distant time and place. Matriculation at the College is taken to signify implicit agreement with the Academic Honor Code, available at: <http://muse.union.edu/honorcode>

It is each student's responsibility to ensure that submitted work is their own and does not involve any form of academic misconduct. Students are expected to ask their course instructors for clarification regarding, but not limited to, collaboration, citations, and plagiarism. Ignorance is not an excuse for breaching academic integrity. Students are also required to affix the full Honor Code Affirmation, or the following shortened version, on each item of coursework submitted for grading:

Union College has an honor code as follows. That material will appear in every notebook you submit to me for grading.

As a student at Union College, I am part of a community that values intellectual effort, curiosity and discovery. I understand that in order to truly claim my educational and academic achievements, I am obligated to act with academic integrity. Therefore, I affirm that I will carry out my academic endeavors with full academic honesty, and I rely on my fellow students to do the same.

Scholastic dishonesty is misrepresenting someone else's work as your own and will not be tolerated.

For this class in particular, we encourage working together during class time. If you missed something, talk to me, talk to your classmates or reach out via piazza or helpdesk. However ALL HOMEWORK must be completed individually. You are encouraged to:

- Talk about concepts in solutions
- Discuss ideas
- Look up online documentation, or examples

You MAY NOT:

- Share code
- Look at another student's code
- Look up solutions to specific problems on the internet
- Copy or paste any text or code to or from a large language model

Ultimately, I may choose to call you into office hours to explain the choices you made in solving a problem. You should be comfortable with explaining the choices you made, how that choice was implemented and why.

Your goal for discussions about any assignment should be that you come away with a better understanding of the problem and of possible ways to approach it so that you can then try out these approaches on your own. You should never leave a discussion with just an answer, without an understanding of how to arrive at that answer. A good general guideline for any

discussion, or interaction with an AI model, is that you should not leave the discussion with anything written or typed.

## **15 CONTACTING ME OUTSIDE OF CLASS**

The best method for contacting me outside of class is to stop by my office during Office Hours or whenever my office door is open. You can also schedule a time with me if my regular office hours don't work. Please contact me via email ([doinggg@union.edu](mailto:doinggg@union.edu)) and we can arrange a phone or zoom call. I respond to emails as soon as possible, but it may take up to one day to get a response during the term, and longer between terms.

## **16 ADDITIONAL RESOURCES**

### *Mental Health and Campus Resources*

Union College is committed to supporting and advancing the mental health and well-being of our students. During the course of their academic careers, students often experience personal challenges that contribute to barriers in learning, such as drug/alcohol problems, strained relationships, chronic worrying, persistent sadness or loss of interest in enjoyable activities, family conflict, grief and loss, domestic violence, difficulty concentrating, problems with organization, procrastination and/or lack of motivation. Students also sometimes come to college with a history of learning difficulties (e.g., any form of special education), experience difficulties succeeding in a particular subject (e.g., math, reading), or have experienced some form of trauma be it emotional or physical (e.g., head injury). These mental health concerns can lead to diminished academic performance and can interfere with daily life activities. If you or someone you know has a history of mental health concerns or if you are unsure and would like a consultation, a variety of confidential services are available. The Eppler-Wolff Counseling Center provides free counseling and therapy services (including psychiatry) to all enrolled students. Please call (518) 388-6161 or visit the Wicker Wellness Center in person any weekday between 8:30 and 5:00 to schedule an initial contact appointment. Visit the Counseling Center website for more information.

In a crisis situation, or after hours, contact Campus Safety at (518) 388-6911. The National Suicide Prevention hotline also offers a 24-hour hotline at (800) 273-8255.

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students ([dos\\_office@union.edu](mailto:dos_office@union.edu)) for support or drop into office hours Monday - Friday 8:30 am - 5:00 pm in the Reamer Campus Center room 306. Furthermore, please notify me if you are comfortable doing so and I will provide any resources I can. The Union College Persistence Fund can provide financial support in times of unexpected hardship to cover



needs such as emergency medical expenses not covered by insurance and basic living expenses. Additional sources of support are outlined on the Dean of Students webpage: <https://www.union.edu/dean-students> .

## **16.1 ADDENDA**

Any community agreed upon additions:

## 17 TENTATIVE SCHEDULE

| W  | TOPIC                               | Due                          | Notes |
|----|-------------------------------------|------------------------------|-------|
| 1  | Scope of Biological Data            | HW 01                        |       |
| 2  | Arc of Analyses                     | HW 02                        |       |
| 3  | Sequence Alignment                  | HW 03, Project Proposal      |       |
| 4  | Data Wrangling                      | HW 04                        |       |
| 5  | Dimensionality Reduction            | Exploratory Analyses         |       |
| 6  | Visualization                       | Figure Drafts                |       |
| 7  | Hypothesis Testing                  | Results                      |       |
| 8  | Peer Review                         | Peer Review                  |       |
| 9  | Methodological Comparative Analysis | Response to Reviewers        |       |
| 10 | Project Presentations               | Final Project & Presentation |       |

## 18 PROGRESS TRACKER

|                                                         | Week | Week | Week | Week | Week | Week | Week | Week | Week | Week |
|---------------------------------------------------------|------|------|------|------|------|------|------|------|------|------|
| Learning Goals                                          | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   |
| How is sequence data transformed into quantitative data |      |      |      |      |      |      |      |      |      |      |
| Data wrangling                                          |      |      |      |      |      |      |      |      |      |      |
| Data visualization                                      |      |      |      |      |      |      |      |      |      |      |
| Interdisciplinary communication                         |      |      |      |      |      |      |      |      |      |      |
| Aquire and prepare data                                 |      |      |      |      |      |      |      |      |      |      |
| Descriptive analysis                                    |      |      |      |      |      |      |      |      |      |      |
| Independent project                                     |      |      |      |      |      |      |      |      |      |      |

# Grading

Table 2: Grading Scheme

| Course Element     | Grade Point Contribution | Notes   |
|--------------------|--------------------------|---------|
| Final Project      | 25%                      | written |
| Final Presentation | 25%                      |         |
| Problem Sets       | 25%                      | written |
| Peer Review        | 25%                      | written |
| total              | 100%                     |         |

Table 3: Letter Grade Scale

| Letter Grade | Percent range |
|--------------|---------------|
| A            | 93 - 100      |
| A-           | 90 - 92       |
| B+           | 87 - 89       |
| B            | 83 - 86       |
| B-           | 80 - 82       |
| C+           | 77 - 79       |
| C            | 73 - 76       |
| C-           | 70 - 72       |
| D            | 60 - 69       |
| F            | 0 - 59        |



# Assignments



*CSC483 HW01*





# *CSC 483: Homework 1, Due Jan 13, 2026*

Rosalind problems 1 - 6: A review of Python.

## *1. Setup Colab Notebook (1 pts)*

1. Make a copy of the Colab Notebook template:

[https://colab.research.google.com/drive/1PdhQTAnEY\\_IPhAQkBxZcACU0gJoOp6oP?usp=sharing](https://colab.research.google.com/drive/1PdhQTAnEY_IPhAQkBxZcACU0gJoOp6oP?usp=sharing)

2. Rename it with your own name: [Name]\_CSC483\_template.ipynb
3. Fill your name into the header. For now, leave the date and assignment name generic.
4. Move this notebook to your class folder on Google Drive, in a well-organized directory.
5. Share it with me, doingg@union.edu
6. Now make a copy of your edited template, save it to your drive, and rename it: [Name]\_CSC483\_HW01.ipynb
7. You will also share your HW01 notebook with me as your submission.

## *2. Rosalind Problems (1 pts)*

Enroll in our class on Rosalind: <https://rosalind.info/classes/enroll/750c73565c/>

Be sure to use your Union email for your account.

You should see the first assignment (problems 1-6).

### *Format*

For each Rosalind problem, use chunks of Markdown text and python code to work out your solution.

Each Rosalind problem will give you a sample data and output to work on. Then, when you are ready, you will download a new dataset and past the corresponding result into the prompt box on Rosalind. I am not asking you to submit code to Rosalind as you will save it in your Colab notebook.

### *Note on reading/writing files*

For some problems you may need to read or write files. In these cases, mount your google drive and upload/save files there.

```
from google.colab import drive
drive.mount('/content/drive/')
```

To check which directory your notebook is mounted to, you can use bash within python chunks

```
!ls
```

You can then use the appropriate relative paths to your files of interest. For example, I used:

```
f = open("drive/MyDrive/Teaching/CSC483-W26/test_text.txt", "r")
```

### *3. Setup Check-in Notebook (1 pts)*

1. Make a copy of the Check-in Notebook template:

<https://colab.research.google.com/drive/1e4a0xjIn9oUVCwZNcZrkv3lM4iyDfSxK?usp=sharing>

2. Rename it with your own name: [Name]\_CSC483\_check-in\_template.ipynb

3. Follow the instructions in the Colab Notebook.
4. Share your notebook with me (doingg@union.edu) as your submission.

#### 4. *Office Hours (1 pts)*

1. After completing your check-in notebook, come to office hours some-time in the next 2 weeks.
  - Location: ISEC 070
  - Student/Office Hours:
    - Wednesday 2:00 pm - 3:30 pm
    - Thursday 4:00 pm - 5:30 pm
    - subject to change, check course website for most up-to-date schedule
    - drop-in or schedule a 15 minute slot: <https://calendar.app.google/8bus6pfDvyphR9ar5>
    - by appointment for another time or over zoom
  - Be careful not to procrastinate, 2 classes worth of students share the office hours.



*CSC483 HW02*



## *CSC 483: Homework 2, Due Jan 20, 2026, at 1 pm (with time for me to make copies before class)*

*1. Rosalind problems 7 - 13: working with sequence data. (2 pts)*

1. Make a copy of your homework colab template, save it to your drive, and rename it: [Name]\_CSC483\_HW02.ipynb
2. You will also share your HW02 notebook with me as your submission.
3. Complete Rosalind problems 7 through 13 in the same format as you used for HW 1.

For each Rosalind problem, use chunks of Markdown text and python code to work out your solution.

Each Rosalind problem will give you a sample data and output to work on. Then, when you are ready, you will download a new dataset and past the corresponding result into the prompt box on Rosalind. I am not asking you to submit code to Rosalind as you will save it in your Colab notebook.

*Work on activity: plotting metadata (1 pts)*

We'll have 30 mins in class on Tuesday before looking at each others plots, so spend some time working on the metadata notebook if you will need more time than that.



Your grade point will come from being ready to present/share after the 30 mins of class time.

1. Also in Colab, keep working with the metadata introduced in class.
2. Make at least 3 plots analyzing or summarizing columns.
3. Write down (in Markdown) a comparison you think would be interesting to make, why you think it would be interesting and what you hypothesize the result would look like.
  - you don't have to succeed in your comparison yet, if you'd need more tools/libraries/functions to do so
  - it is more important that you have a reason for making the comparison and an estimate to which you could compare your results

### *3. Update Check-in Notebook (1 pts)*

1. Review your check-in notebook and scores for last week's homework assignment
2. Include a Markdown chunk that attests to my notes or requests a correction/update

*CSC483 HW03*



## *CSC 483: Homework 3, Due Jan 27, 2026, at 1 pm (with time for me to make copies before class)*

*1. Rosalind problems 14 - 20: sequence & alignment. (3 pts + 4 pts of PR)*

1. Make a copy of your homework colab template, save it to your drive, and rename it: [Name]\_CSC483\_HW03.ipynb
2. You will also share your HW03 notebook with me as your submission.
3. Complete Rosalind problems 14 through 20 in the same format as you used for HW 1.

Note that for this set, some problems direct you to use web-based programs rather than writing your own python code.

For each Rosalind problem, use chunks of Markdown text and python code to work out your solution.

Each Rosalind problem will give you a sample data and output to work on. Then, when you are ready, you will download a new dataset and past the corresponding result into the prompt box on Rosalind. I am not asking you to submit code to Rosalind as you will save it in your Colab notebook.

4. In-class peer review holds an additional 4 points in the grading scheme, so be sure to finesse your notebook for easy reading.

*2. Update Check-in Notebook (1 pts)*

1. Move your reflections from HW1, HW2 and HW3 into your check-in notebook.
2. Review your check-in notebook and scores for last week's homework assignment
3. Include a Markdown chunk that attests to my notes or requests a correction/update

*CSC483 HW04*



## *CSC 483: Homework 4, Due Feb 3, 2026, at 1 pm (with time for me to make copies before class)*

*1. Rosalind problem 20: sequence & alignment. (3 pts + 4 pts of PR)*

1. Make a copy of your homework colab template, save it to your drive, and rename it: [Name]\_CSC483\_HW04.ipynb
2. You will also share your HW04 notebook with me as your submission.
3. Complete Rosalind problem 20 in the same format as you used for HW 1.

Note that scoring matrices can be loaded via a python library:

```
!pip install scoring-matrices
from scoring_matrices import ScoringMatrix
blosum62 = ScoringMatrix.from_name("BLOSUM62")
blosum62['A', 'C']
```

4. In-class peer review holds an additional 4 points in the grading scheme, so be sure to finesse your notebook for easy reading.



*2. Update Check-in Notebook (1 pts)*

1. Move your reflections from HW1, HW2 and HW3 into your check-in notebook.
2. Review your check-in notebook and scores for last week's homework assignment
3. Include a Markdown chunk that attests to my notes or requests a correction/update

*CSC483 HW05*



## *CSC 483: Homework 5, Due Feb 10, 2026, at 1 pm*

*1. Begin your project notebook (2 pts + 2pts in PR)*

1. Make a copy of the project guidelines:

<https://colab.research.google.com/drive/1xqpoZnPbjISUX3gsE59iw5CO-XQ6nNc?usp=sharing>

2. Read through the guidelines, but then you can delete the markdown cells with instructions, they are available on the course website.
3. Load whichever dataset and meta-data from `first_data` or `second_data` you are most interested in. Careful of data sizes!
4. Add an exploratory plot with a similar level of finesse as we plotted in class (as in it does not need to be perfect), like a PCA or t-SNE.
5. Add a markdown cell with some notes doing your best to describe why and how we are making PCA and t-SNE plots.

Note: This may seem repetetive to what we did in class, and feel free to copy and paste your own work from notebook to notebook. In starting a new notebook you will have the chance to create a more organized analysis that combines elements of `first_data` and `second_data` without some of the extra bits I added as examples. Most notably, you only have to load the datasets you are actually examining rather than the multiple meta-data files

and multiple RNA-seq files I've loaded in `first_data` and `second_data`. You do not need to complete all of the following, but for reference, recall the in-class instructions from the `second_data` notebook:

### *Decision points*

1. use smaller rnaseq datasets ( filtered 6k x 16k) OR use gene P/A
2. reduce dim of genes OR dim of samples (transpose or not to transpose)
3. connect to metadata of samples OR of genes
  - red by genes means plotting samples and vv
  - merge rnaseq with sample metadata, use SRX numbers
  - merge gene p/a with annotation, use `first_name_comp`
  - merge rnaseq with gene annotation use `first_name_comp`
4. Choose a color scheme
5. Does DR technique separate out groups, as seen by color?

### *Mini-steps*

1. color all points on PCA NOT blue
2. color half the points one color, half another
3. color the *first* sample a unique color
4. load metadata (as we did in `first_data`)
5. turn `pca` from numpy array to `pd df`
6. figure out rownames for PCA `df`, based on column names `df4`

### *2. Update Check-in Notebook (1 pts)*

1. Move your reflections from HW4 into your check-in notebook.
2. Review your check-in notebook and scores for all previous HW assignments.
3. Include a Markdown chunk that attests to my notes or requests a correction/update.

*CSC483 HW06*



## *CSC 483: Homework 6, Due Feb 17, 2026, at 1 pm*

### *1. Project Proposal (2 pts + 2pts in PR)*

1. Continue working in your notebook from last week, adding any exploratory analyses you do to come up with a question to answer for your project.
2. Include what data you will use to answer your question and what fields you will focus on.
3. Specify how you will filter your data to be able to answer your question without loading unnecessary data.
4. Specify which dimensionality reduction method(s) you would like to use. These can be any [flavor of PCA](#) or [embeddings like t-SNE](#) or any other method: [UMAP](#), [ICA](#), [NMF](#) etc.
5. Describe how you might need to group or merge your data to answer your question.
6. End by writing out questions you are still confused about. This doesn't need to be a full reflection, but come to class ready to ask questions that will help you get your project rolling.

### *Decision points*

1. use smaller rnaseq datasets ( filtered 6k x 16k) OR use gene P/A
2. reduce dim of genes OR dim of samples (transpose or not to transpose)



3. connect to metadata of samples OR of genes

- red by genes means plotting samples and vv
- merge rnaseq with sample metadata, use SRX numbers
- merge gene p/a with annotation, use first\_name\_comp
- merge rnaseq with gene annotation use first\_name\_comp

4. Choose a color scheme

5. Does DR technique separate out groups, as seen by color?

*CSC483 HW07*



## *CSC 483: Homework 7, Due Feb 24, 2026, at 1 pm*

### *1. Project Draft (5 pts )*

1. In a **new** Colab notebook, using your previous Colab notebook as an outline/guide, conduct your analyses. Come to class ready for peer review. This will not be the final draft you are graded on, but it will be a chance for feedback from your peers and me before you submit the draft that will be graded.
  - reference the project rubric as you go - this is a very flexible assignment
  - remember to cite any references
  - also think about which highlights of your results you will use for your presentation
  - if you are feeling stuck, utilize my office hours, the librarian consultation services (<https://www.union.edu/schaffer-library/ask-a-librarian>), and/or each other! Just include visits/conversations in your references.
2. End by writing out questions you were unable to answer. This doesn't need to be a full reflection, but might help you pivot your project.

### *Decision points*

1. use smaller rnaseq datasets ( filtered 6k x 16k) OR use gene P/A
2. reduce dim of genes OR dim of samples (transpose or not to transpose)
3. connect to metadata of samples OR of genes

- red by genes means plotting samples and vv
- merge rnaseq with sample metadata, use SRX numbers
- merge gene p/a with annotation, use first\_name\_comp
- merge rnaseq with gene annotation use first\_name\_comp

4. Choose a color scheme
5. Does DR technique separate out groups, as seen by color?

*CSC483 HW08*



## *CSC 483: Homework 8, Due March 3, 2026, at 1 pm*

### *1. Project Draft (3 pts + 5pts in PR + 5 pts PR)*

1. Make substantial progress on your project. Come to class ready for another round of peer review (using the project rubric) followed by a **practice presentation**. This will be your last chance for feedback from your peers and me before you submit the draft that will be graded.

Be sure to include *at least* the following sections:

- + introduction to your focus/question
- + methodological description of what data you are using and how you will process it
- + results as plots and quantitative summaries/statistics

2. End by beginning your reflection on what you think your project's strengths and weaknesses are. Be sure to address both the project product as well as the process you went through to create it.





*CSC483 HW09*



# *CSC 483: Assignment 9, Due March 12th 2026*

Note that the due date of this assignment is **BEFORE** your presentation

## *1. Project Check-in (5 pts)*

1. Meet with me for 15 minutes for a final project practice. Use *either* of the following calendars to schedule a time:
  - Office Hours: <https://calendar.app.google/rSeYBUayMo1XpsAk9>
  - Extra times: <https://calendar.app.google/h8EsK88uQiqT9f8j9>
  - Note: if we have recently met in office hours and you feel totally prepared, you may add a markdown cell to your check-in notebook indicating as such and receive these points

## *2. Sign up for a presentation time slot*

- [Google Sheet with time slots](#)

## *3. Add your slide(s) to the common slideshow*

- [CSC483 Google Slides](#)



# *Final Project*

## *Project Details*

The goal of the final project is to explore an unruly set of biology data and find a story to tell. You may use the strain-level gene data (DNA), the gene expression data (RNA) and/or the sample annotations (meta-data). With whatever data you choose, find an interesting question to answer that challenges:

- your analytic skills (using Python to get, manipulate, analyze and visualize)
- your story telling skills (how do you use the above, to tell a coherent story using data)

BOTH are equally valued.

Here are some tips / thoughts about specifically what I need to see:

1. The data are not already pre-processed or perfectly packaged so you will need to perform some (but not all) of merging, joining, creating new columns, selecting columns, changing types of data. These operations are likely best done using the Pandas library.
2. To tailor your data to your question/story, you will need to decide criteria for including/excluding data – both rows (e.g. genes or samples)

and columns (e.g samples, strains or meta-data feilds). It is impossible to give minimum or maximum requirements on data size. However, too little data ( less than 500 instances) leaves little chance for finding useful patterns. Too few attributes (columns, less than 5) means that you either need to merge in complementary data, or create your own additional columns. Conversely too much data (in rows or columns) means that you almost certainly have to START reducing the size of your data for realistic processing on Colab, even in a high-mem runtime.

3. You will need to create visualizations. But, don't just graph everything because you can, think about what plots you can show that best assist with your story.
4. I would like some specific analytics – discussion in combination with analytic approaches – i.e. what features are important for a specific problem, what trends exist in the data and how do you explain them? I'm not looking for you to be right, I'm looking for your story to be well-motivated by data
5. I cannot tell you how long your notebook should be. HOWEVER consider the size of the grade which this notebook is worth. It should contain paragraphs of text that could serve as sections of a paper (introduction, methods, results, discussion) in addition to the plots and tables.
6. I am looking to learn. Part of your grade will be in how well YOU take ME on a journey through data which honestly, at the end, you are likely to find something I never would have. Telling that story is a combination of code (show me results, do not comment out the good stuff) and the write up. This IS part essay, part presentation, part coding exercise.

### *Rubric*

- maximum points: 12
  - 6 for analysis
  - 4 for storytelling
  - 2 for reflection

|                                 | Exceeding expectations (2 pts)                                                                                                                                                                                                                                                                                                      | Arriving at expectations (1.5 pts)                                                                                                                                                                                | Not meeting Expectations (0.5 pts)                                                                                                                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Analytical /<br>Mathematic      | <p>Broad use of Pandas such as merging, joining, creating new columns, selecting columns, changing types of data.</p> <p>Visualizations are chosen carefully in order to convey multiple levels of information.</p> <p>Data selection is thoughtful and considers rows and columns or multiple data types (DNA, RNA, meta-data)</p> | <p>Some use (1-2 instances) of Pandas such as selecting data or summarizing columns.</p> <p>Some visualizations</p> <p>One type of data is used, but it is “large” enough to support conclusions</p>              | <p>No data manipulation using Pandas and/or data sets are left as loaded.</p> <p>Visualizations don’t contribute to the story in clear ways.</p> <p>Data is left unfiltered or is too restricted to support conclusions.</p> |
| Communication /<br>Storytelling | <p>Writing builds a convincing story by explaining analysis results, providing interpretation and strong evidence that the data is understood.</p> <p>Analysis asks and answers a clearly motivated question.</p>                                                                                                                   | <p>There’s a story but it is missing pieces: results are partially explain, insufficient evidence that data was understood</p> <p>A question is posed but no clear answer or larger implications are conveyed</p> | <p>No discernible story; demonstrates a lack of engagement with the data</p> <p>No question is posed.</p>                                                                                                                    |



|            | Exceeding expectations (2 pts)                                                                                                                                                | Arriving at expectations (1.5 pts)                                                                                                                                      | Not meeting Expectations (0.5 pts)                                                                                      |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Reflective | Student thoughtfully identified specific strengths and areas for improvement. A complete picture of the student's work process with a plan for greater success in the future. | Student identified strengths or areas for improvement but lacked specificity or sufficient development in either area. Only a partial picture of the student's process. | Reflection lacked considerations and areas of improvement, did not incorporate or address suggestions from peer review. |

### *Presentation Rubric*

- maximum points: 12
  - 6 for analysis
  - 4 for storytelling
  - 2 for reflection

|                                 | Exceeding expectations (2 pts)                                                                         | Arriving at expectations (1.5 pts)                                            | Not meeting Expectations (0.5 pts)                                                                         |
|---------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Analytical /<br>Mathe-<br>matic | Convinving analysis accepting or rejecting a null hypothesis with statistical / quantitative evidence. | Use of quantitative descriptions but no clear answer to the central question. | Data does not supoprt the conclusions or, no quantitative results presented, no testable hypothesis posed. |

|                              | Exceeding expectations (2 pts)                                                                                                    | Arriving at expectations (1.5 pts)                                                                                                    | Not meeting Expectations (0.5 pts)                                                           |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|                              | Visualizations are selected from results carefully and are highly informative                                                     | Some visualizations presented but not clear highlights of the project                                                                 | Visualizations are uninformative, misleading or don't contribute to the story in clear ways. |
|                              | Data selection is clearly justified (DNA, RNA, meta-data)                                                                         | Data selection is stated by not defended                                                                                              | Data selection is unclear.                                                                   |
| Communication / Storytelling | <p>Presentation has a clear and convincing arc presented in time.</p> <p>There is a clear central question which is answered.</p> | <p>There's a story but it lacks a resolution or does not fit the time frame.</p> <p>A question can be gleaned but is not central.</p> | <p>Presentation is not focused on a story.</p> <p>No question is made explicit.</p>          |
| Reflective                   | Student responds to questions thoughtfully providing new information when needed.                                                 | Student responds to questions to the best of their ability but is lacking sufficient breath to engage in discussion.                  | Responses to questions are unclear or do not address the questions asked.                    |



## Notes



*Week 1*

*Week 1 slides*

**CSC 483 - Applied Biological Data Science -  
W1D1**

## Today (105 min)

- Introductions? (15')
- Fermi questions (30')
- What is biological data science? (30')
- Course goals, arc, project (15')
- Colab rules & grading scheme (15')



## Free write: scale of “biology” to “data science”

- what classes or experiences have you had? where do they fall on the spectrum?
- where would you place yourself on the spectrum?
- in which direction do you hope to move by taking this class?

Me:

- Georgia Doing, PhD
- she/her/hers
- computational microbiologist studying human skin microbiome and uncharacterized genes
- interests: computational microbiology, small cells & big data

## My dog Ginny



\* Please email me ahead of office hours if you would like me to make sure Ginny won't be there, not a problem at all!

## You:

- Name and pronouns
- Help me find you on my roster
- What is something you bring to this class? What classes or experiences have you had? Where would you place yourself on the spectrum from biology to data science?
- What is something you hope to learn in this class? In which direction (toward biology or data science) do you hope to move by taking this class?

# Fermi questions

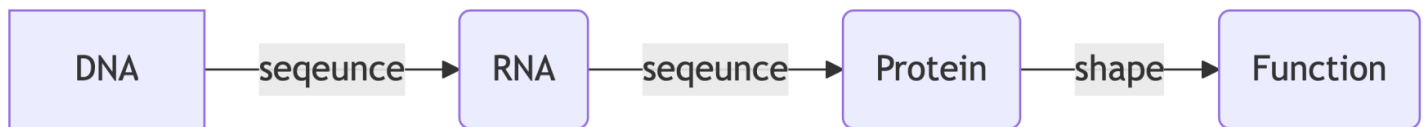
<https://fermi-questions.andrecheek.com/>

- in pairs
- if you want a different question, refresh page
- 10 mins
- answers are in orders of magnitude

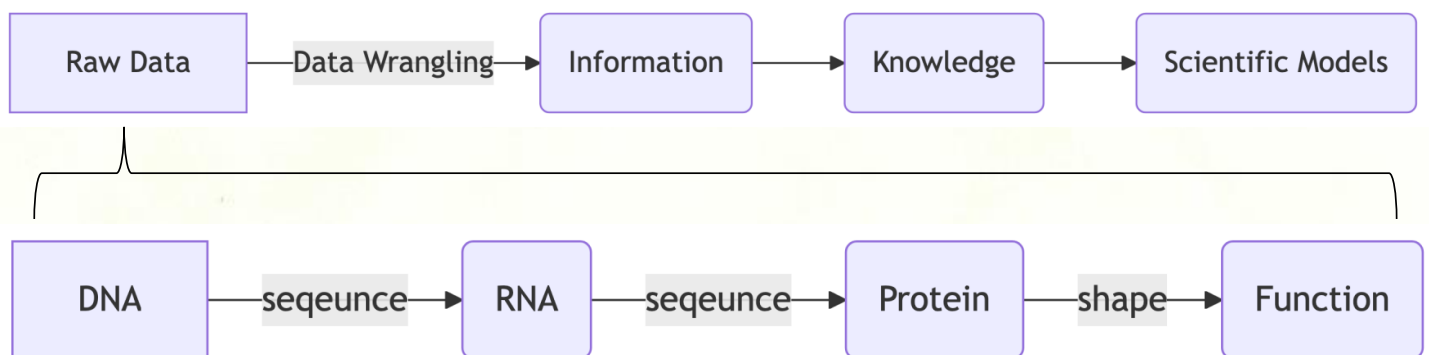
# What is Data Science?



# What is Biological Data?



# What is Biological Data Science?



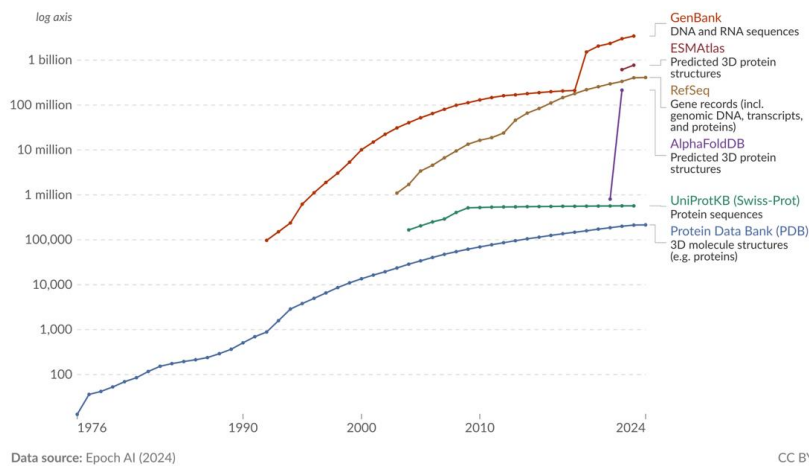


# Biological Data Accumulation

## Number of entries in biological sequence databases

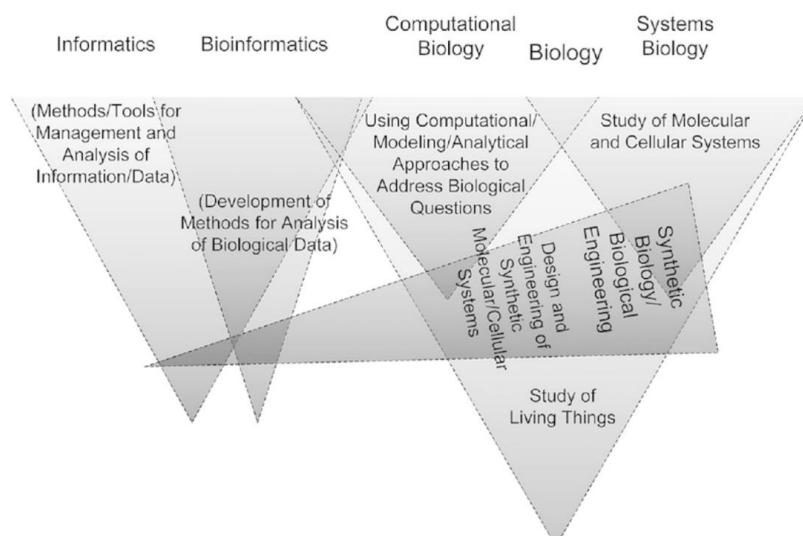
Biological sequence databases store data such as DNA, RNA, and amino acid sequences and 3D protein structures. This data includes entries from GenBank, RefSeq, PDB, UniProtKB/SwissProt, as well as predicted protein structures in AlphaFoldDB and ESMAtlas.

Our World  
in Data



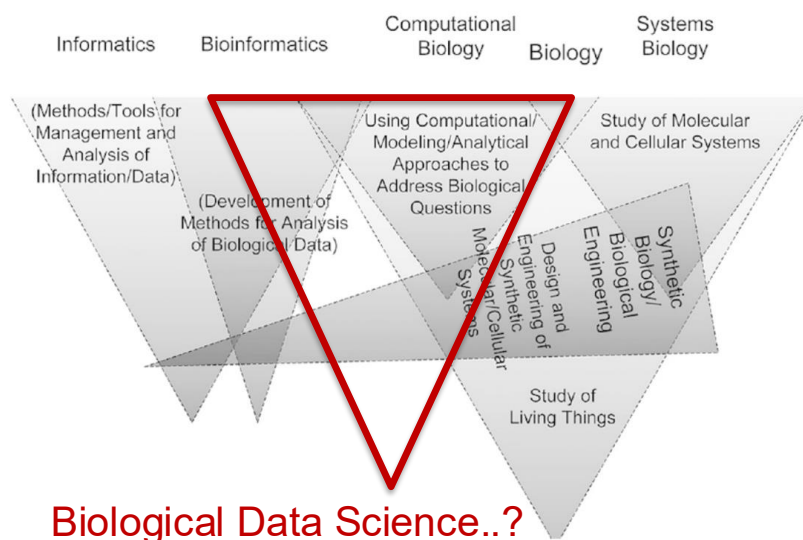
<https://ourworldindata.org/grapher/number-of-entries-in-biological-sequence-databases>

# What is Biological Data Science?



**FIGURE 2.1** A disciplinary map of “informatics,” “bioinformatics,” “computational biology,” “biology,” and “systems biology” according to one practitioner. (Christopher Burge, “Introduction to Computational and Systems Biology” (7.91), lecture, Massachusetts Institute of Technology, February 8, 2007. Reproduced with permission.)

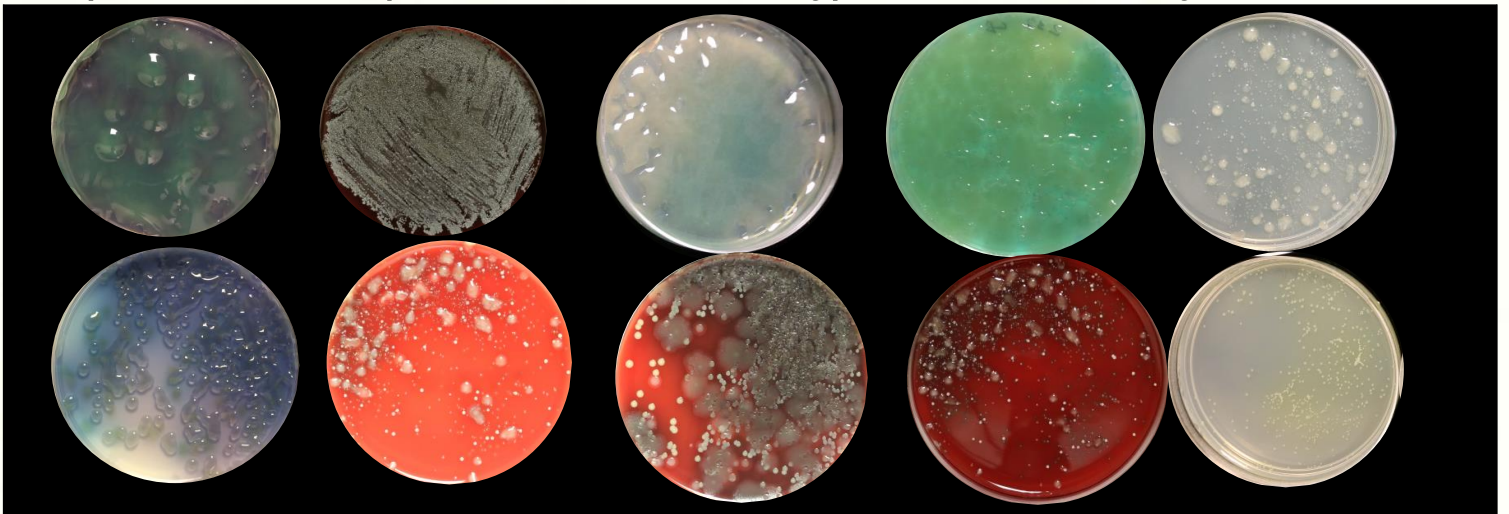
# What is Biological Data Science?



**FIGURE 2.1** A disciplinary map of “informatics,” “bioinformatics,” “computational biology,” “biology,” and “systems biology” according to one practitioner. (Christopher Burge, “Introduction to Computational and Systems Biology” (7.91), lecture, Massachusetts Institute of Technology, February 8, 2007. Reproduced with permission.)

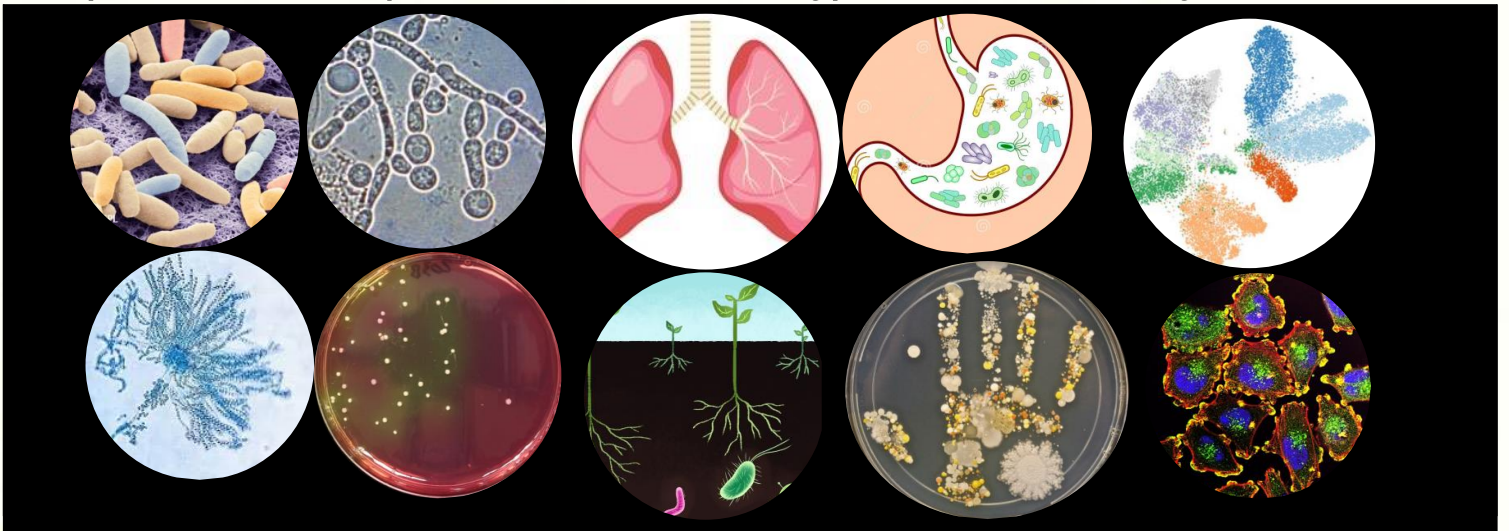
# Biological Data Science to Me

My interests: computational microbiology, small cells & big data



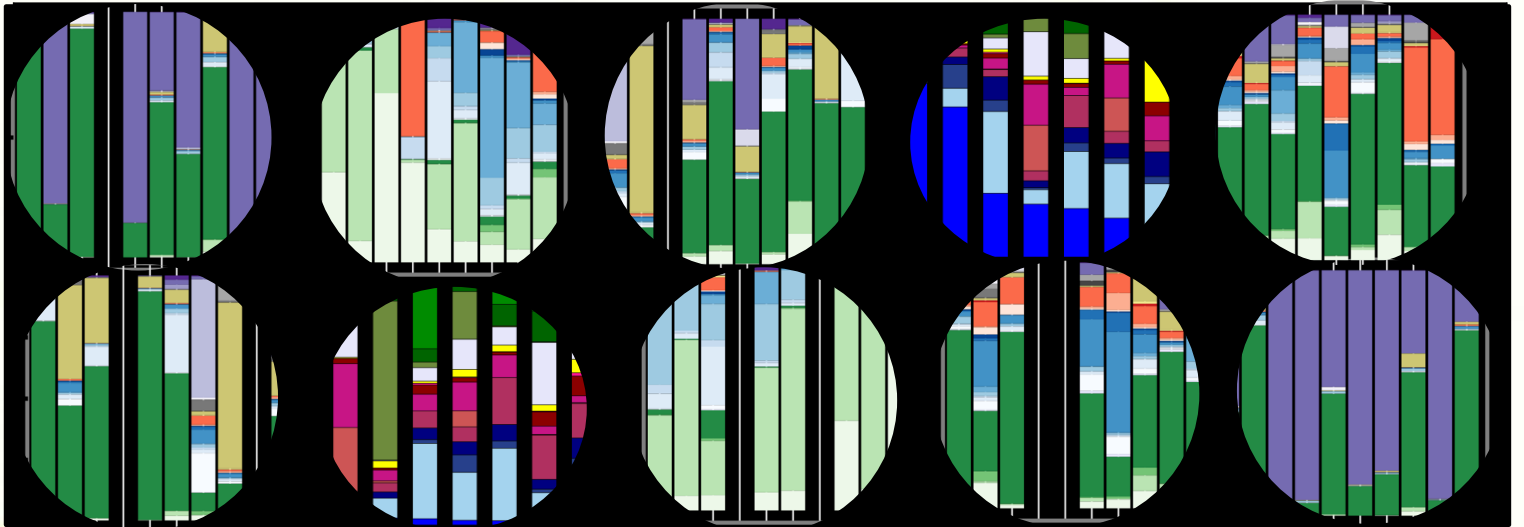
# Biological Data Science to Me

My interests: computational microbiology, small cells & big data



# Biological Data Science to Me

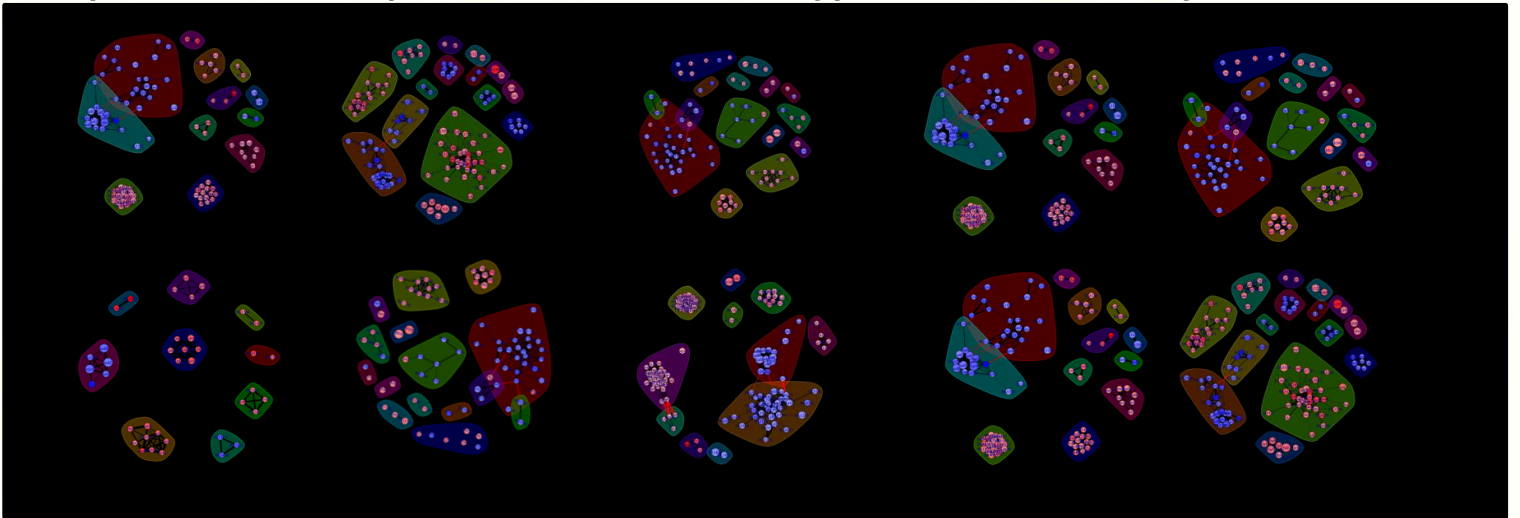
My interests: computational microbiology, small cells & big data





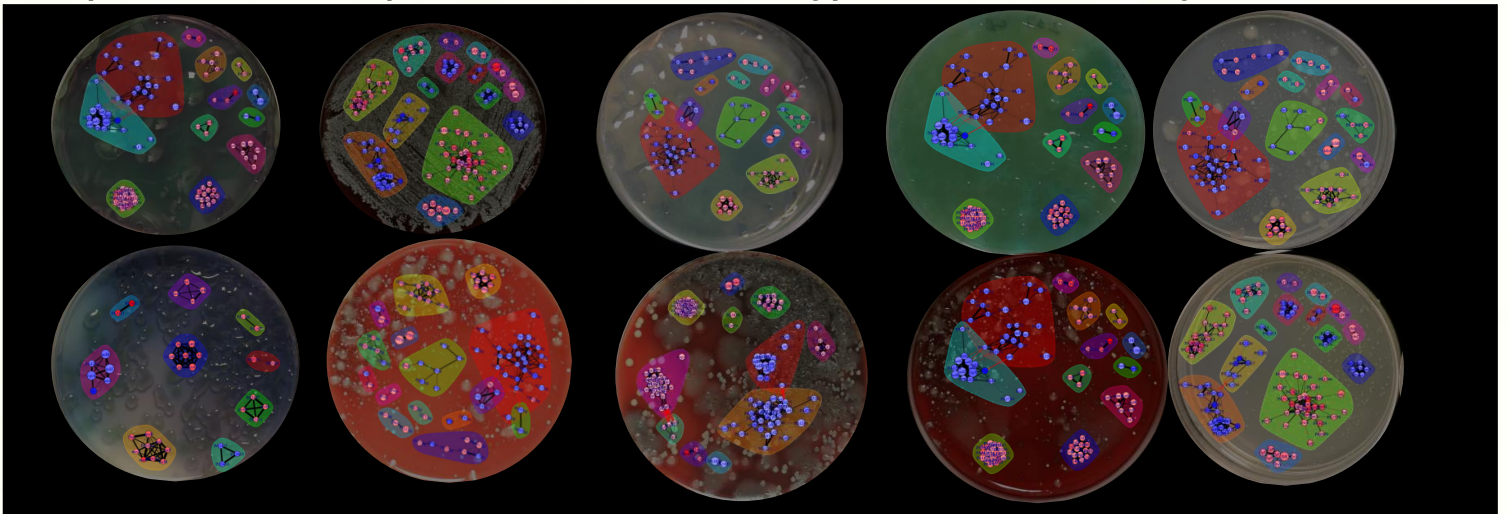
# Biological Data Science to Me

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# Biological Data Science to Me

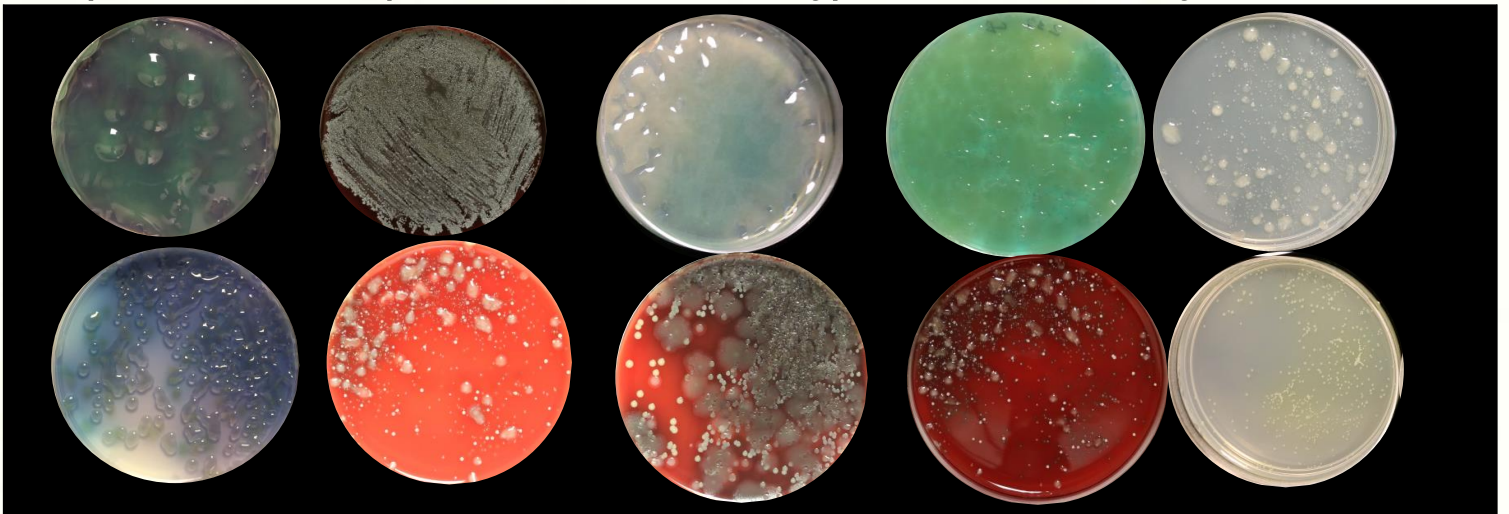
My interests: computational microbiology, small cells & big data





# Biological Data Science to Me

My interests: computational microbiology, small cells & big data



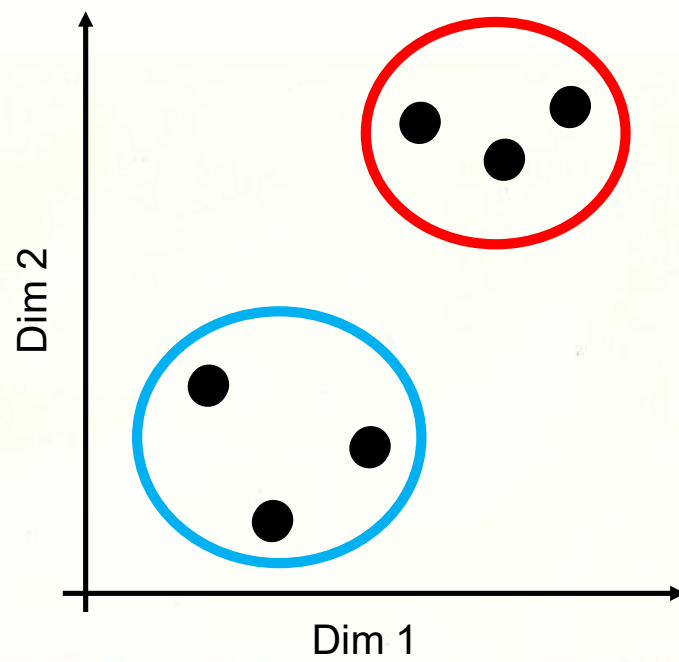
# Biological Data Science to Me

5,549 Genes

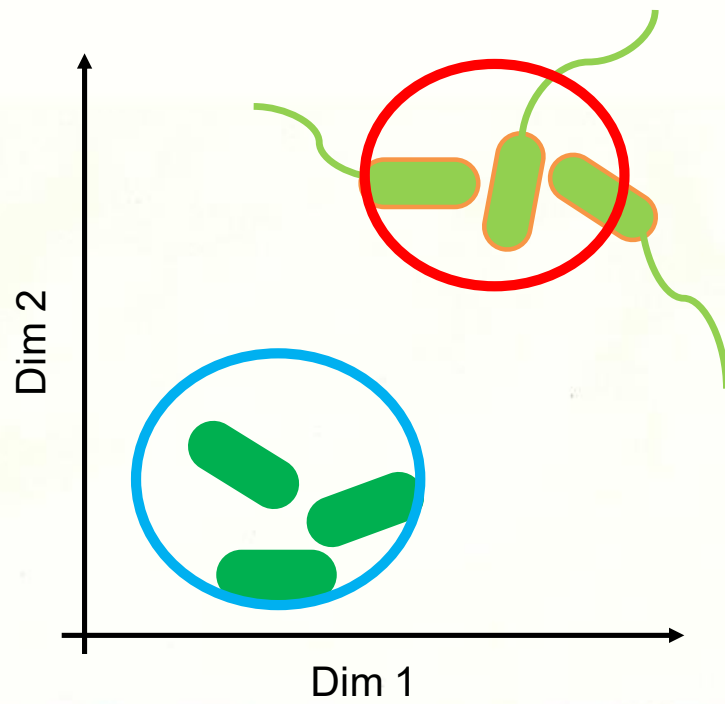
3,011 Samples



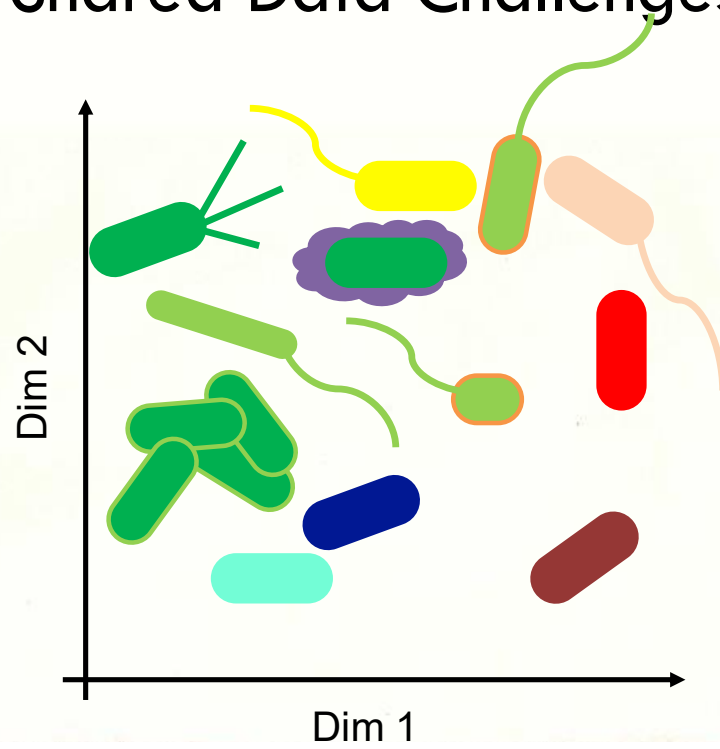
## Generic Data Problems



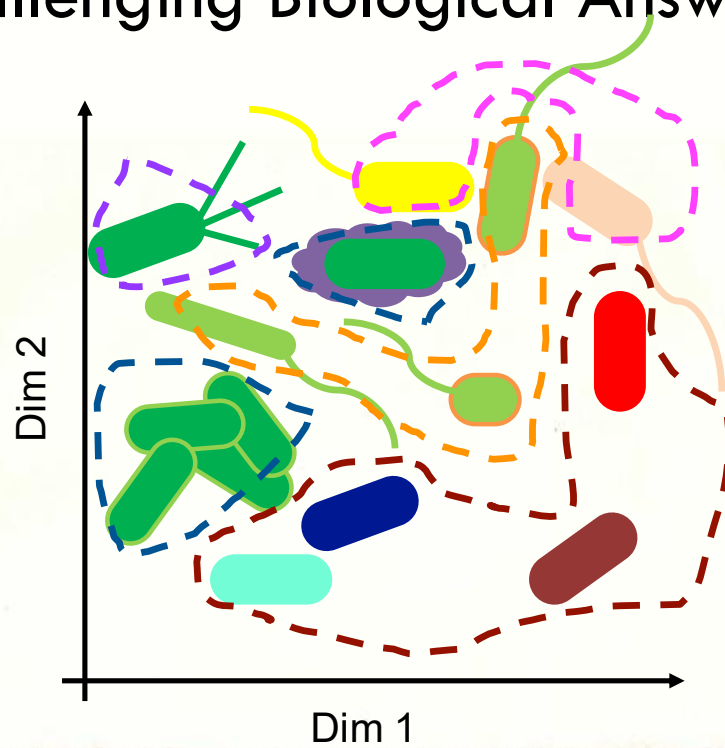
## Specific Biological Questions



## Shared Data Challenges

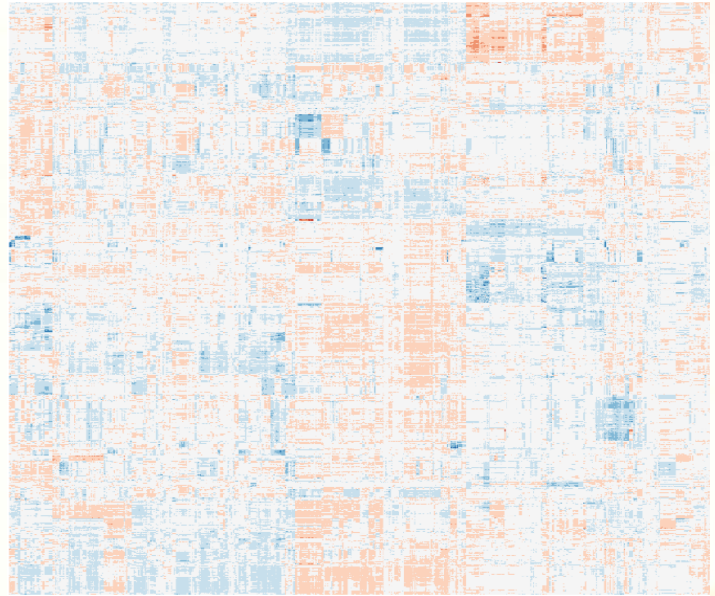
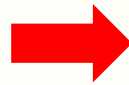
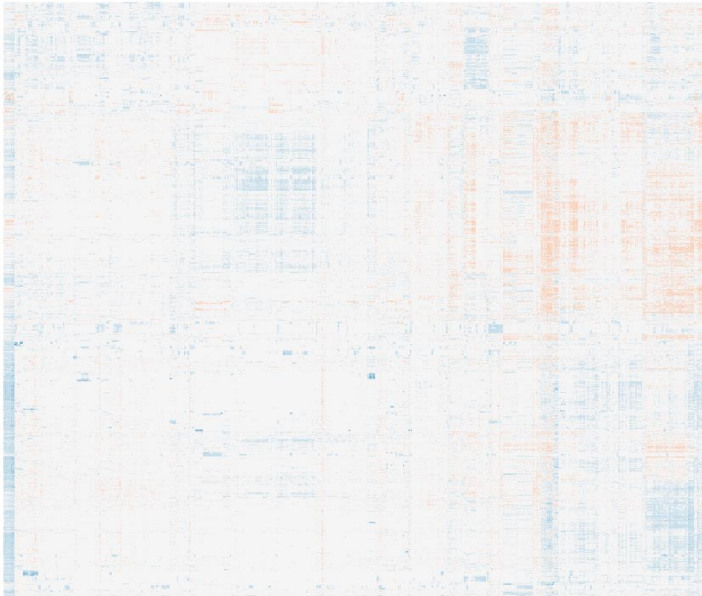


# Challenging Biological Answers





## State-of-the-Art ... Open Questions



## Lab machines: getting started

- Log on:
  - Your Union usernames
  - Default password: the word union followed by your ID number; e.g., union12345

**Always log off before leaving the lab**





## *Week 2*

### *E coli metadata activity*

- colab notebook: [https://colab.research.google.com/drive/1X18-sF7Spht9ZXZlDkY2S\\_zvgEEf4xJG?usp=sharing](https://colab.research.google.com/drive/1X18-sF7Spht9ZXZlDkY2S_zvgEEf4xJG?usp=sharing)
- metadata 1: [https://drive.google.com/file/d/1BB7M3eWo8zPuxmlwSKMzE\\_RBeYrjv1n0/view?usp=sharing](https://drive.google.com/file/d/1BB7M3eWo8zPuxmlwSKMzE_RBeYrjv1n0/view?usp=sharing)
- metadata 2: [https://drive.google.com/file/d/18\\_PkR\\_sOLsQshSM4O1n2hVOcZoxgn8E0/view?usp=sharing](https://drive.google.com/file/d/18_PkR_sOLsQshSM4O1n2hVOcZoxgn8E0/view?usp=sharing)
- metadata 3: [https://drive.google.com/file/d/1\\_cJGV1gwHXzgUe7DMwHVI28nhJcma2tt/view?usp=sharing](https://drive.google.com/file/d/1_cJGV1gwHXzgUe7DMwHVI28nhJcma2tt/view?usp=sharing)

### *Week 2 slides*

# CSC 483 - Applied Biological Data Science

## W2D2

# CSC 483: Applied Biological Data Science

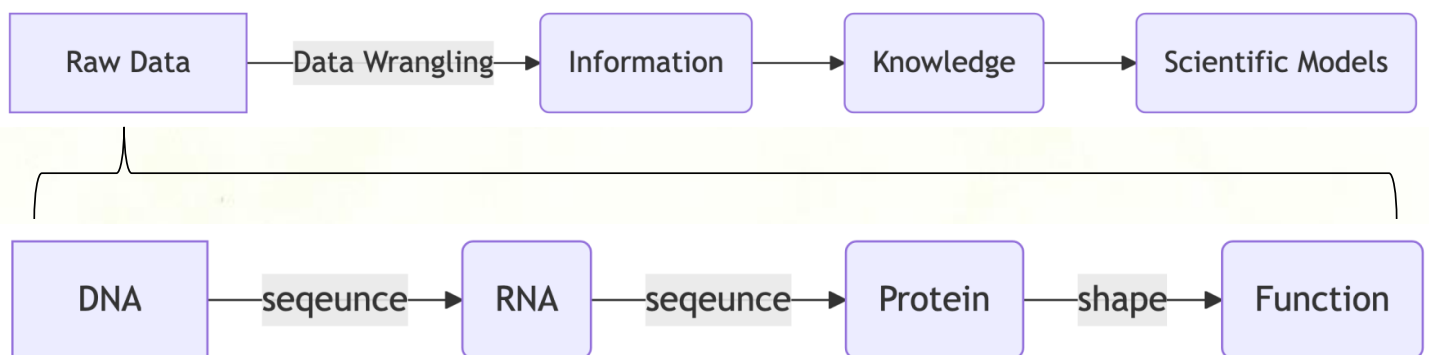
- Georgia Doing, PhD (she/her/hers)
- email: [doingg@union.edu](mailto:doingg@union.edu)
- office: Steinmetz 108B
  - office hours:
    - Wednesday 2:00-3:30 pm
    - Thursday 4:00-5:30 pm

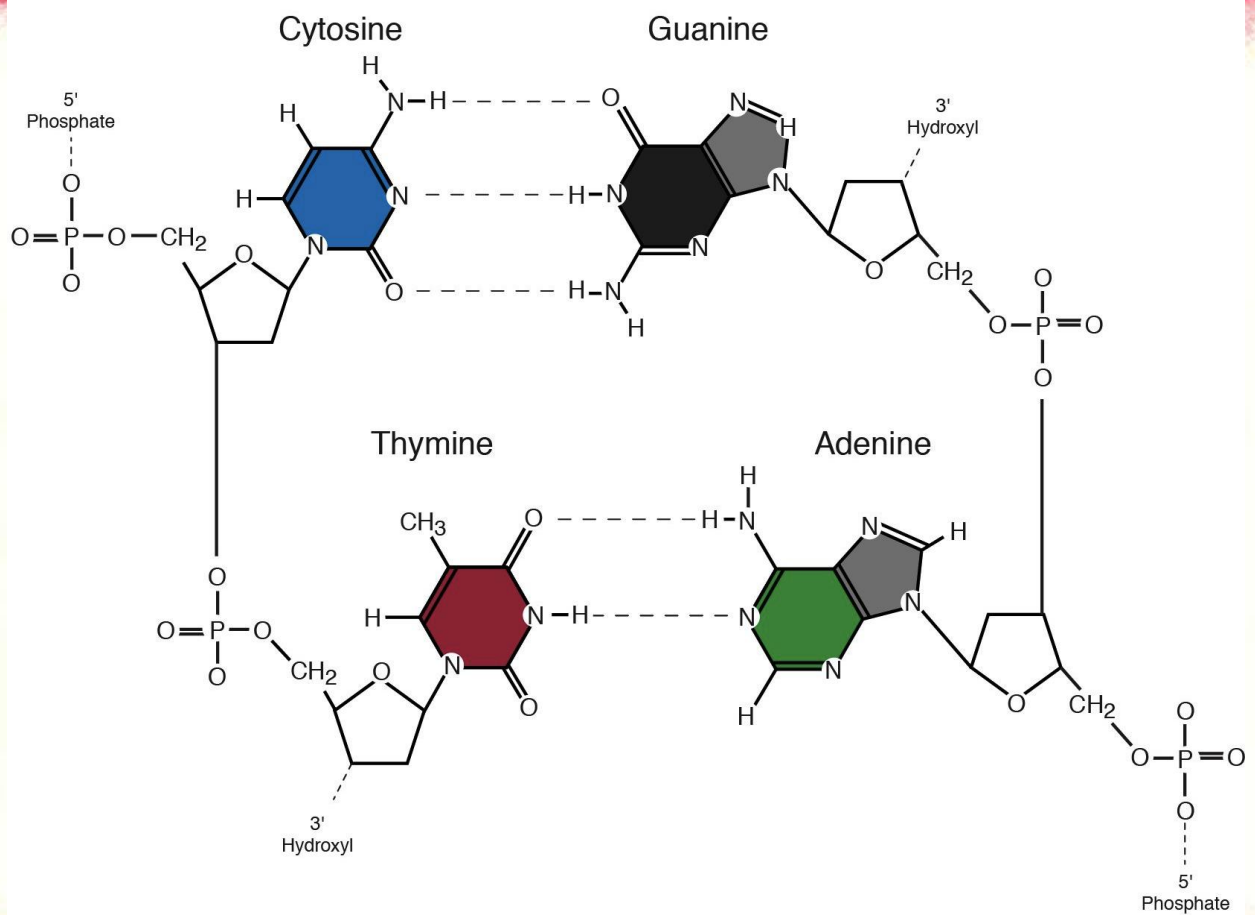
Feel free to drop in whenever my office door is open or schedule an appointment.

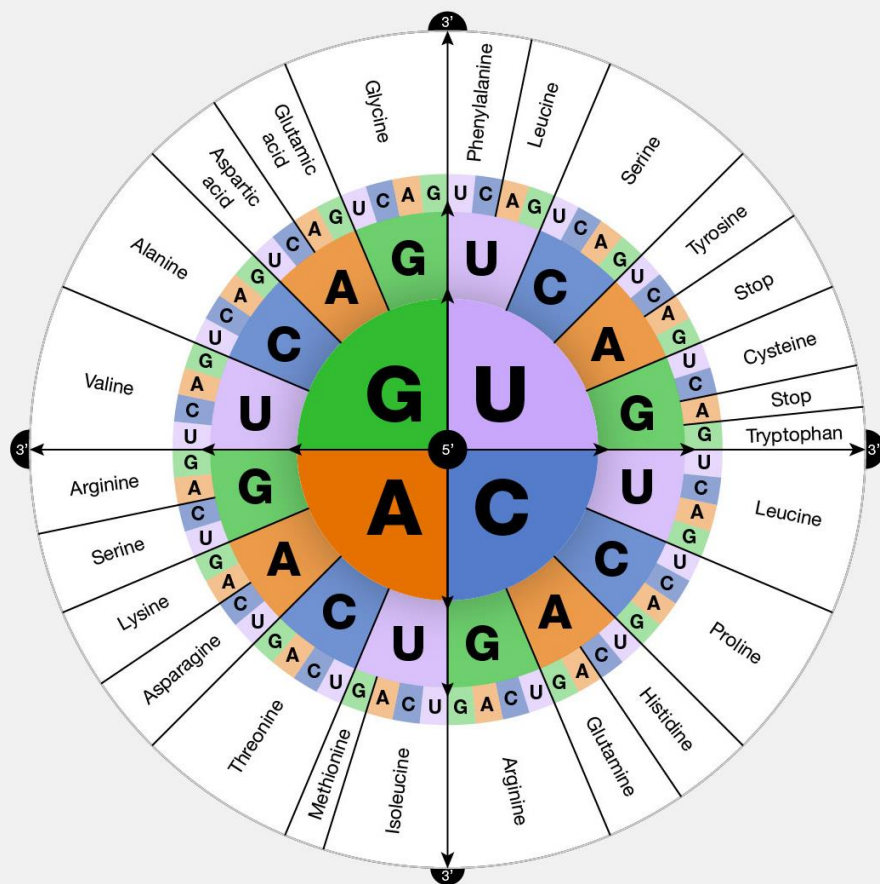
## Today (145 min)

- Paper on methods/background
- Metadata summaries
- Explore data ,brainstorm ideas
- HW:
  - rosalind problems on sequence analysis (w/ colab notebooks)
  - due: tues

# What is Biological Data Science?

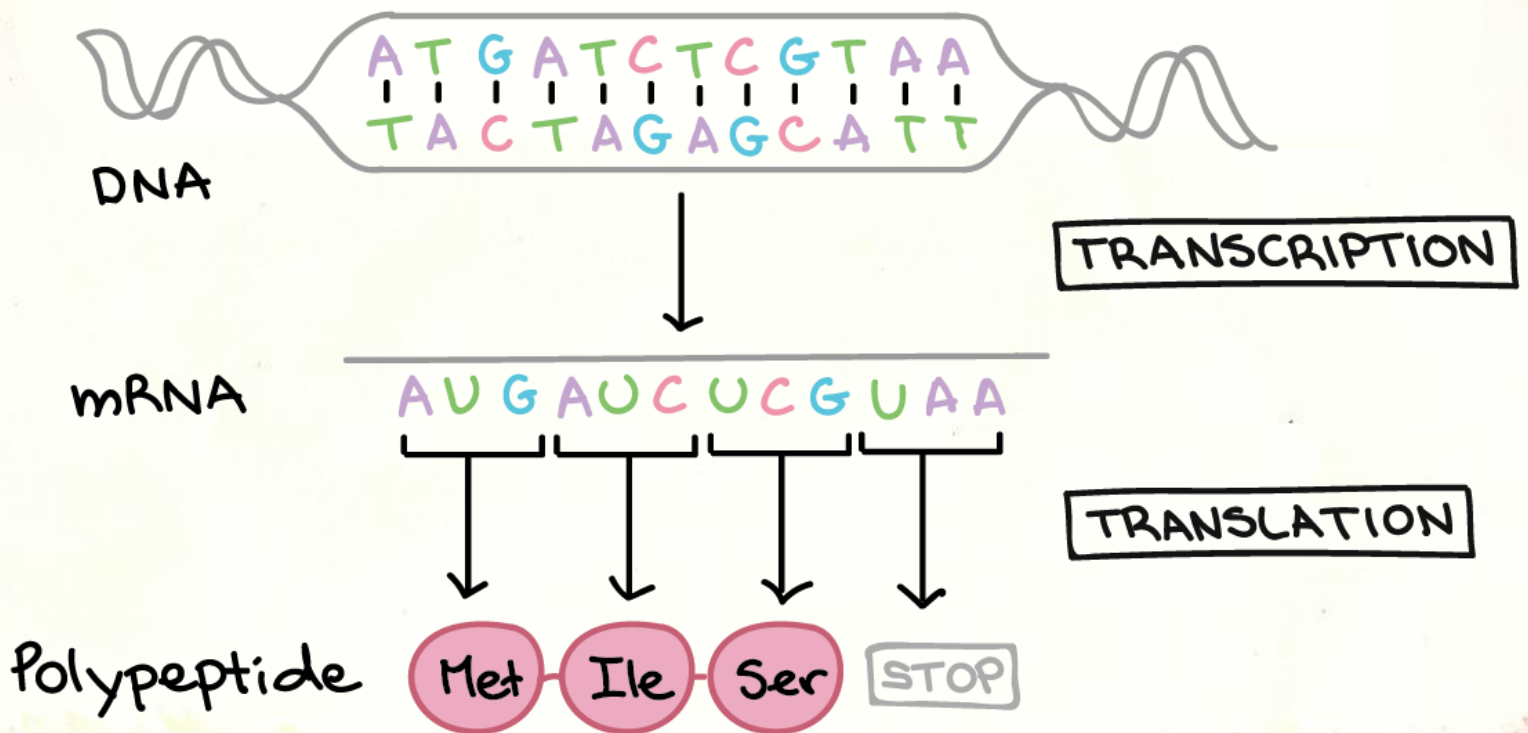




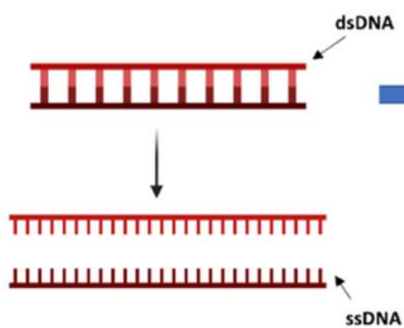




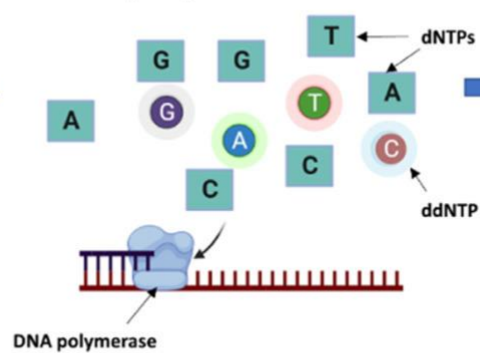
## THE CENTRAL DOGMA



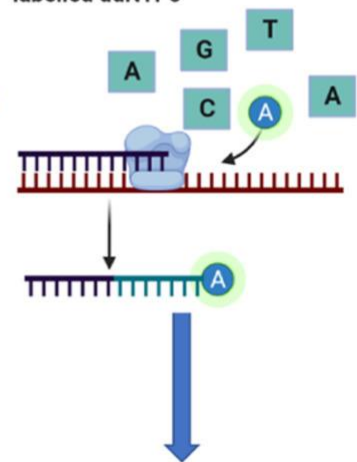
- ① Denaturation of dsDNA into ssDNA template through heat treatment



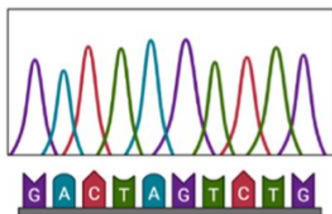
- ② Primer annealing and extension by the DNA polymerase by addition of contemporary dNTPs



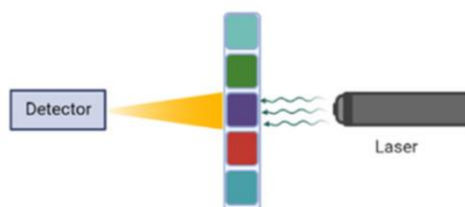
- ③ Termination of each round of primer extension by the fluorescently labelled ddNTPs



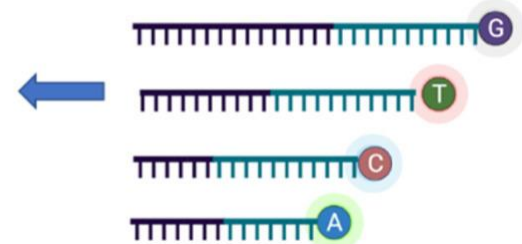
- ⑥ Sequence analysis and reconstruction



- ⑤ Separation of chain-terminated oligonucleotides using gel electrophoresis, preferably single capillary gel electrophoresis



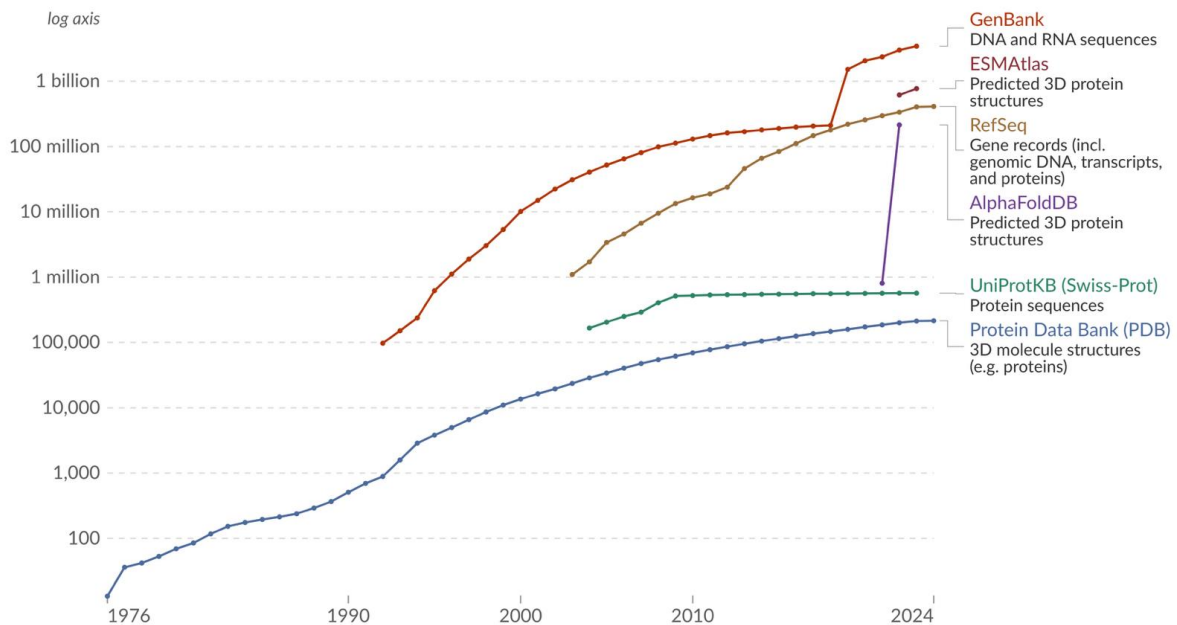
- ④ Fluorescently labelled DNA sample



## Number of entries in biological sequence databases

Our World  
in Data

Biological sequence databases store data such as DNA, RNA, and amino acid sequences and 3D protein structures. This data includes entries from GenBank, RefSeq, PDB, UniProtKB/SwissProt, as well as predicted protein structures in AlphaFoldDB and ESMAtlas.



Data source: Epoch AI (2024)

CC BY

<https://ourworldindata.org/grapher/number-of-entries-in-biological-sequence-databases>

# IGSR: The International Genome Sample Resource

Supporting open human variation data

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Search IGSR



## The International Genome Sample Resource

The 1000 Genomes Project created a catalogue of common human genetic variation, using openly consented samples from people who declared themselves to be healthy. The reference data resources generated by the project remain heavily used by the biomedical science community.

The International Genome Sample Resource (IGSR) maintains and shares the human genetic variation resources built by the 1000 Genomes Project. We also update the resources to the current reference assembly, add new data sets generated from the 1000 Genomes Project samples and add data from projects working with other openly consented samples.



### Variation Consortium, Phase 3

A Human Genome Structural Variation Consortium (HGSVC), funded by NHGRI, have built on their earlier work published in 2010 and 2011 employing multiple technologies for structural variation discovery and the most recent data generated by HGSVC.

A phase of the HGSVC is in progress and expects to add additional human genomes using more complete phased assemblies including longer and more accurate long-read sequence data.

A core set of validated data sets is based on the HGSVC HGSVC-1000.

| Download the list | Download the list                           |
|-------------------|---------------------------------------------|
| Sample            | Populations                                 |
| HG001997          | Admixed African, Yoruba, The Gambia - Mende |
| HG001998          | Mende in Sierra Leone                       |
| HG002177          | Daan in Nigeria                             |
| HG001999          | Colombian in Medellin, Colombia             |
| HG002000          | Peruvian in Lima, Peru                      |
| HG002001          | UK-Liverpool Span in the UK                 |
| HG002002          | Indian Bengali in the UK                    |
| HG002003          | Albanian in Kosovo                          |
| HG002004          | Yoruba in Ibadan, Nigeria                   |
| HG002005          | Japanese in Tokyo, Japan                    |
| HG002006          |                                             |

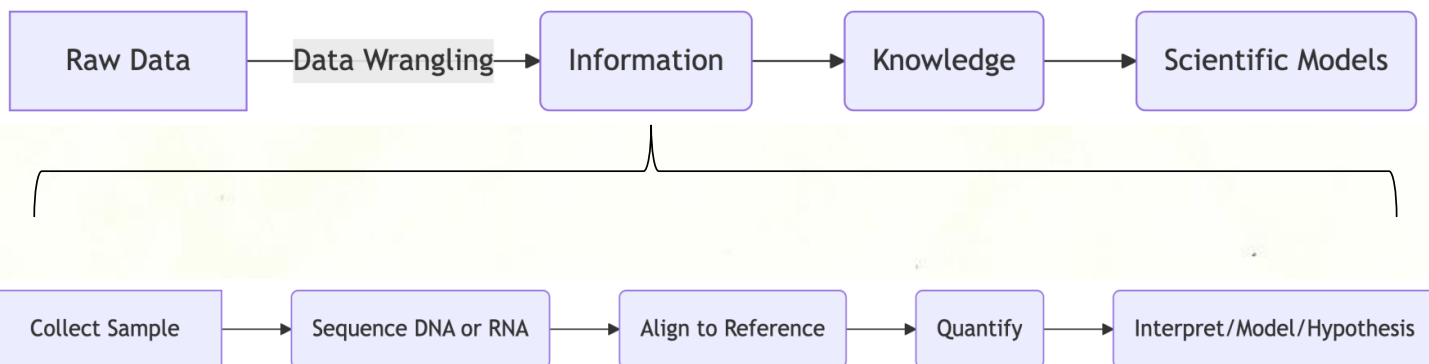
### Access HGSVC data

|     |                                                       |                                                      |                  |
|-----|-------------------------------------------------------|------------------------------------------------------|------------------|
| AFR | 0: 0.706 (20071)<br>A: 0.007 (120)<br>G: 0.903 (1184) | GIK: 0.036 (1342)<br>A: 0.018 (12)<br>G: 0.982 (545) | ANG: 0.167 (104) |
| ACB | A: 0.151 (30)<br>G: 0.849 (165)                       | A: 0.042 (4)<br>G: 0.958 (77)                        | ANG: 0.219 (21)  |
| ASW | A: 0.292 (58)<br>G: 0.708 (94)                        | A: 0.066 (6)<br>G: 0.934 (97)                        | ANG: 0.328 (20)  |
| ESN | A: 0.096 (13)<br>G: 0.904 (185)                       | A: 0.012 (1)<br>G: 0.988 (87)                        | ANG: 0.111 (11)  |
| MSL | A: 0.079 (17)<br>G: 0.921 (186)                       | ANG: 0.047 (6)                                       | GIK: 0.903 (81)  |
| YRI | A: 0.079 (17)<br>G: 0.921 (186)                       | ANG: 0.157 (17)                                      | GIK: 0.843 (81)  |
| AMR | A: 0.385 (93)<br>G: 0.615 (447)                       | A: 0.147 (81)<br>G: 0.853 (447)                      | ANG: 0.436 (181) |
| CLM | A: 0.482 (88)<br>G: 0.518 (110)                       | A: 0.033 (31)<br>G: 0.967 (20)                       | ANG: 0.457 (43)  |
| MXL | A: 0.305 (38)<br>G: 0.695 (69)                        | A: 0.084 (8)<br>G: 0.916 (27)                        | ANG: 0.422 (27)  |
| PEL | A: 0.252 (43)<br>G: 0.747 (127)                       | ANG: 0.059 (5)<br>GIK: 0.940 (47)                    | ANG: 0.388 (33)  |

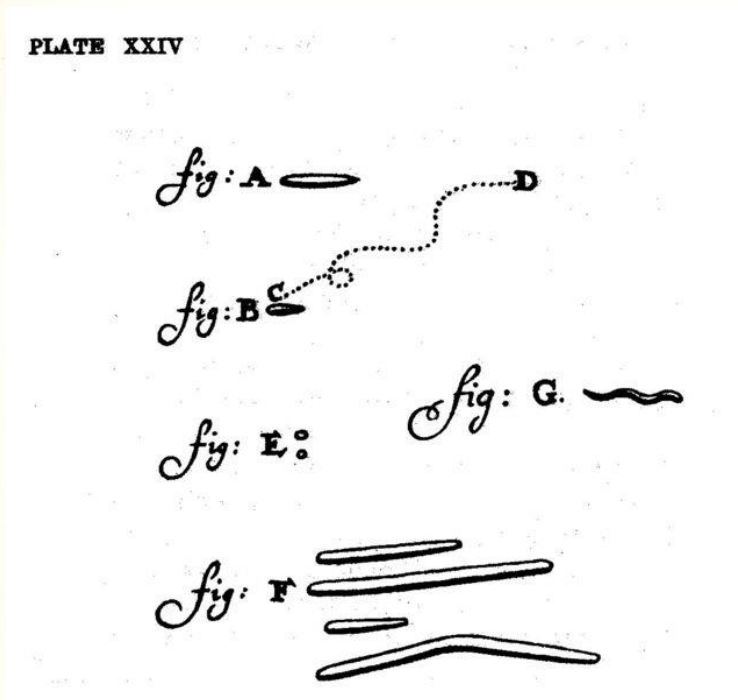
### View variants in genomic context in Ensembl

1000s Genomes Project: <https://www.internationalgenome.org/>

# Experimental Workflow

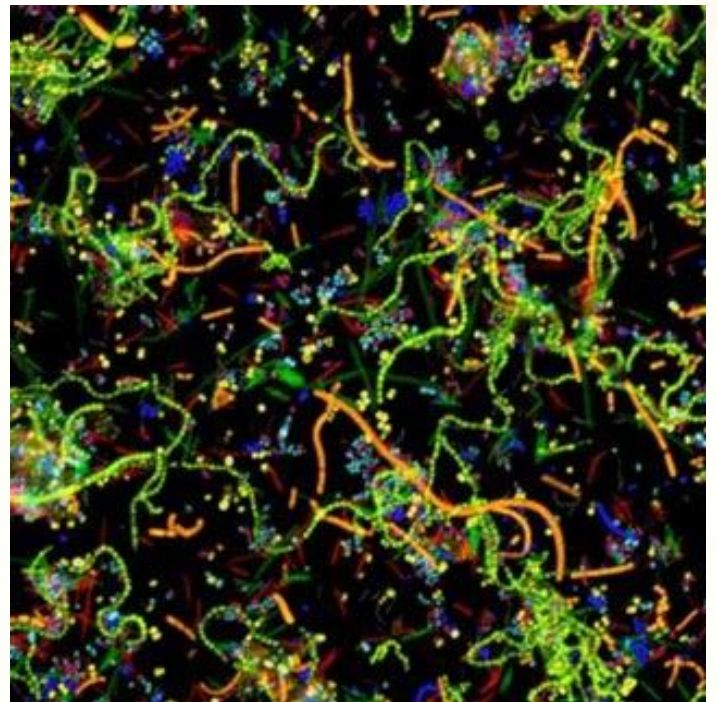


## Oral 'animacules' (1676)



A. van Leeuwenhoek

## Oral microbiome (2011)



Valm et al., PNAS



## Isolate



Genomics / Transcriptomics

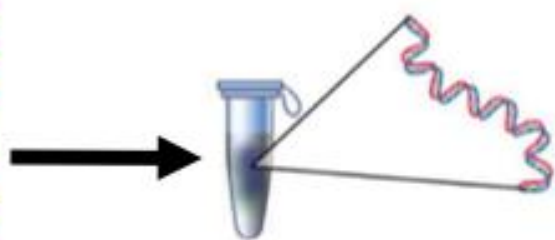
## Community



Metagenomics / Metatranscriptomics



Microbial Community

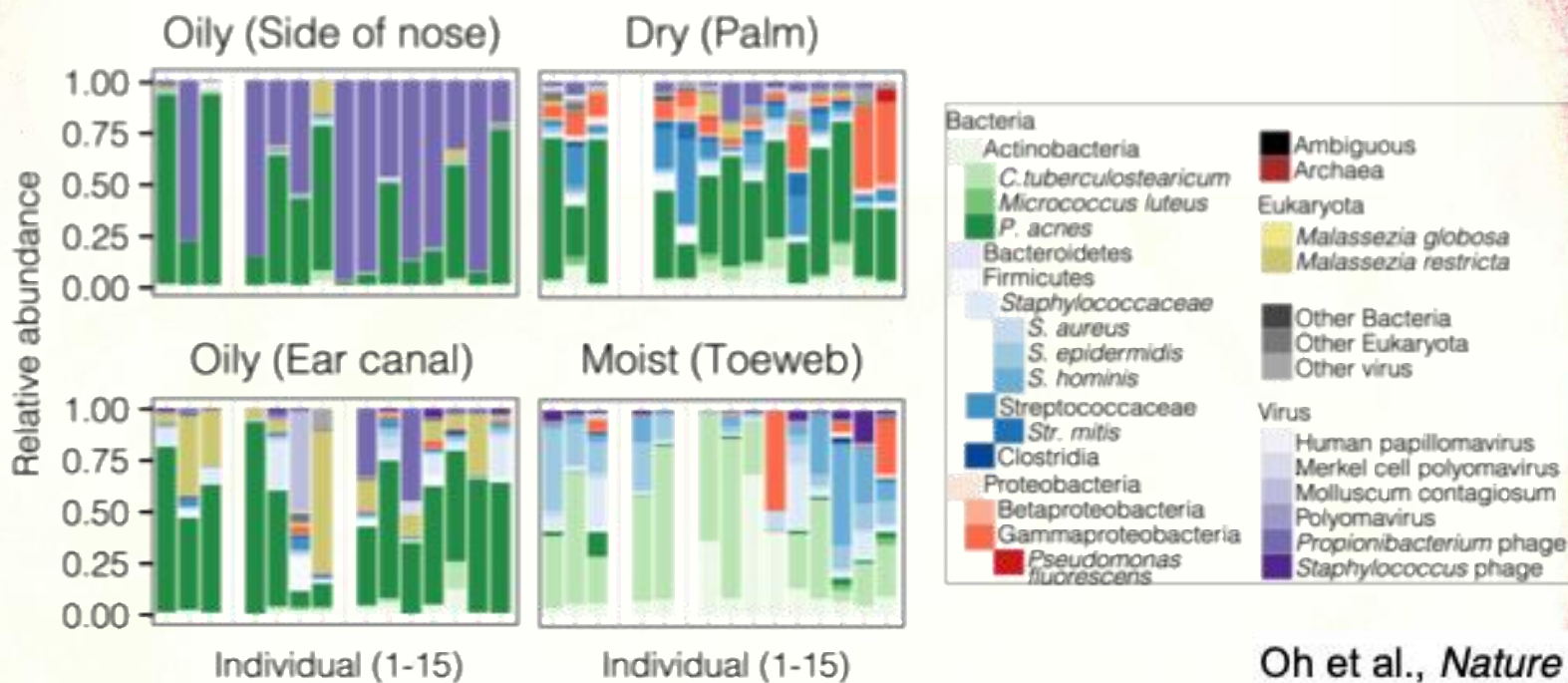


Molecular Extraction



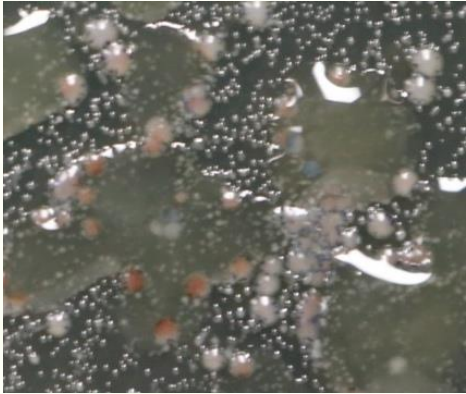
Sequencing



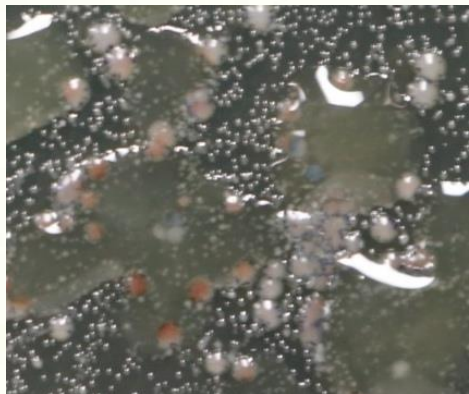


Oh et al., *Nature*

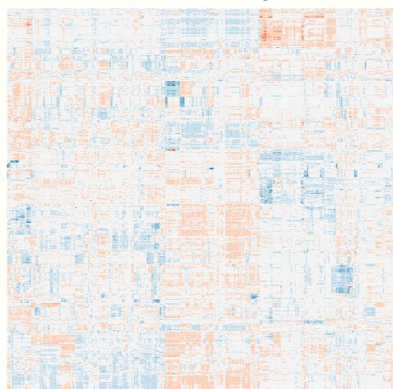
## Microbial Culture



Microbial Culture

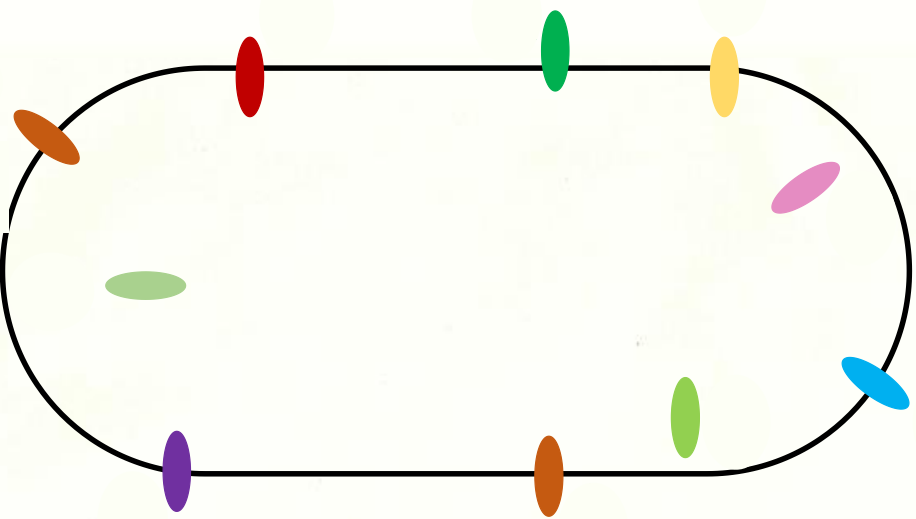


RNA-seq Data



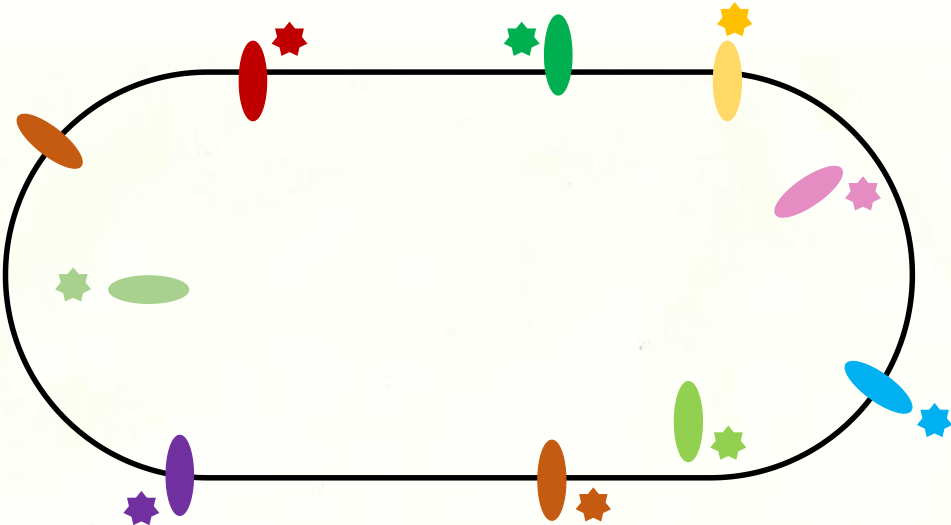


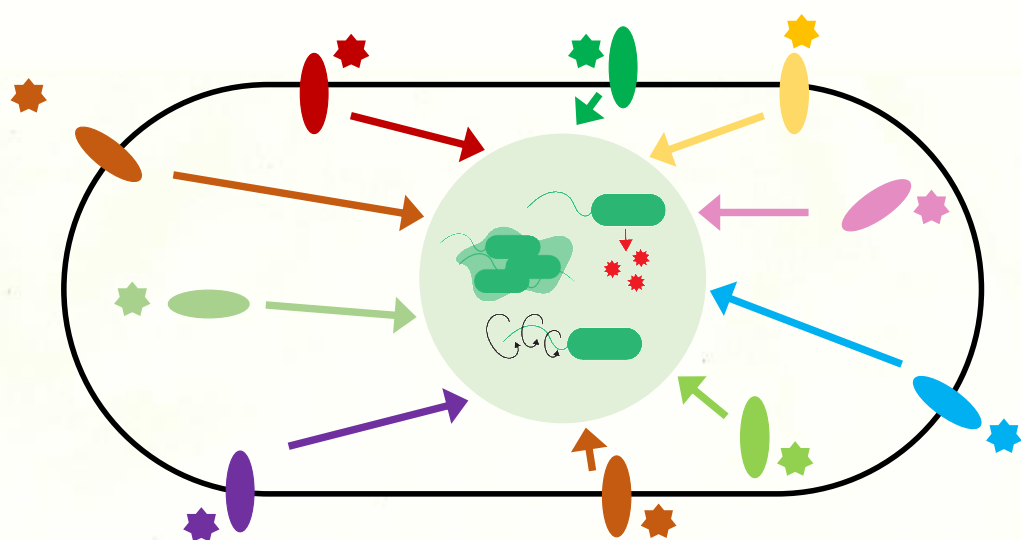
Proteins  
sensors

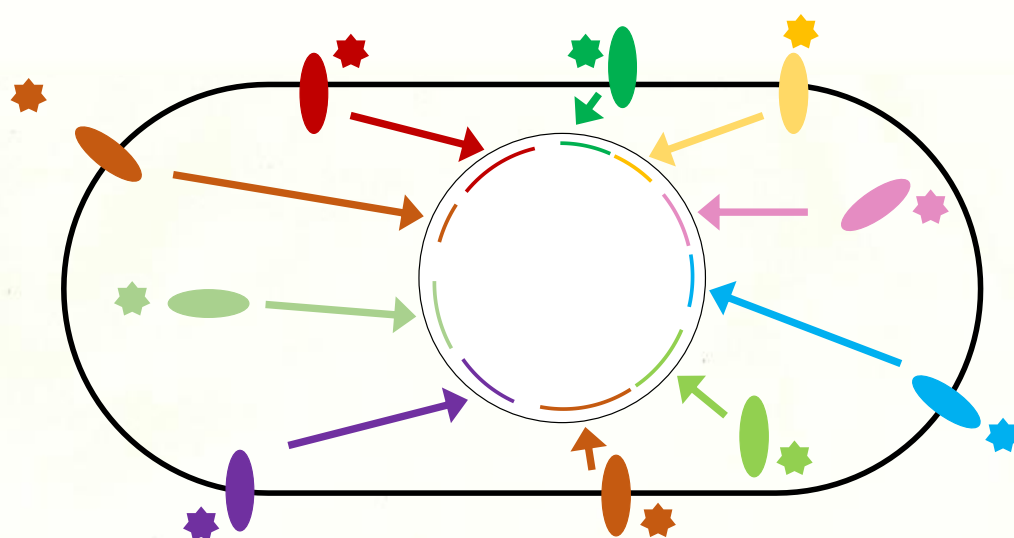


Molecular cues

Proteins sensors

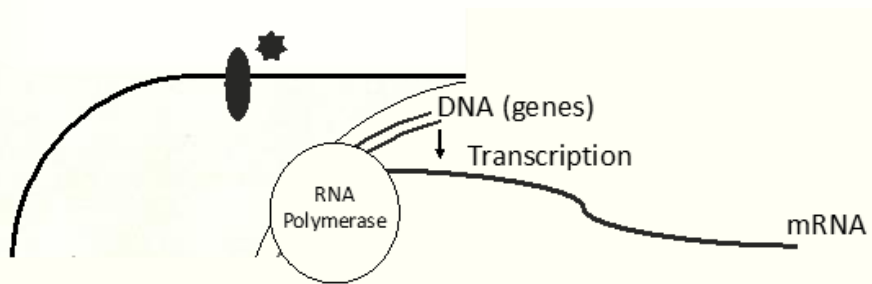


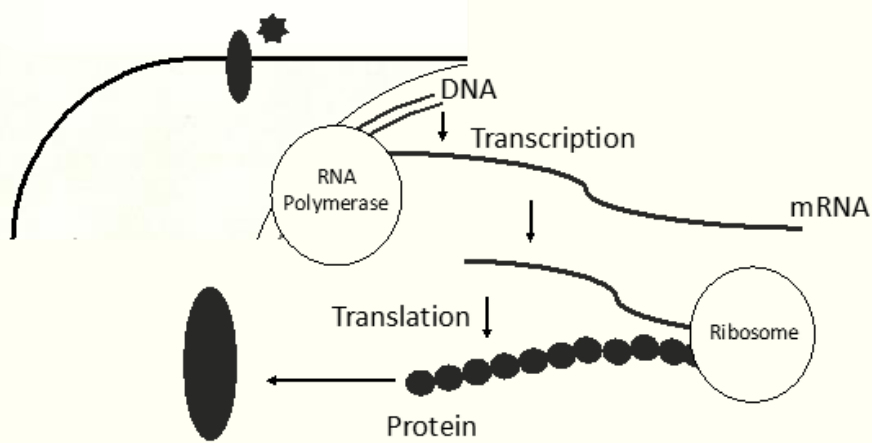


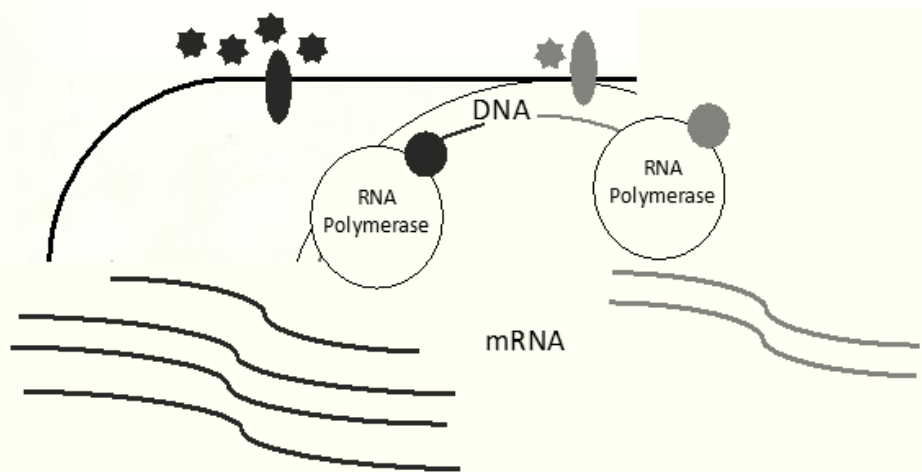


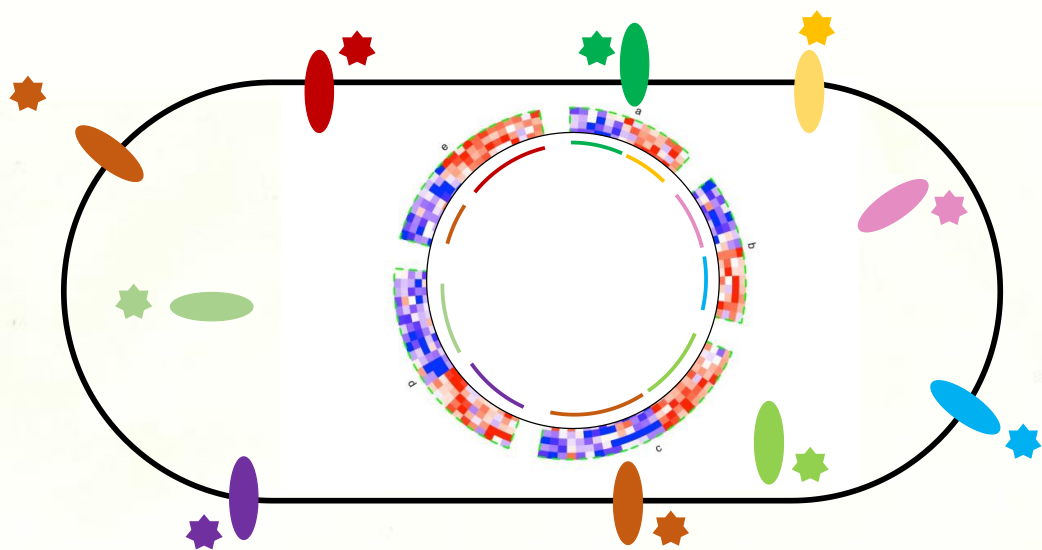


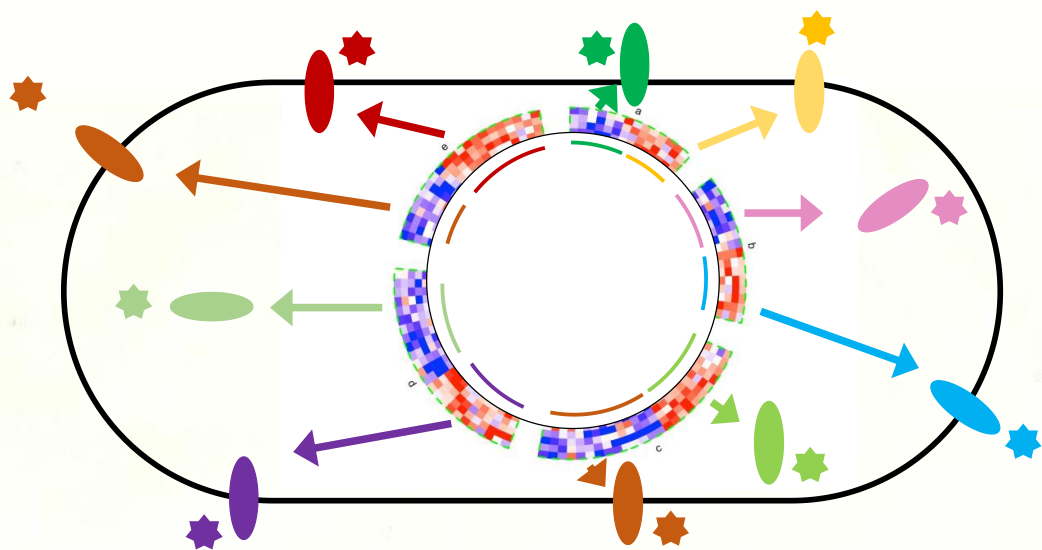


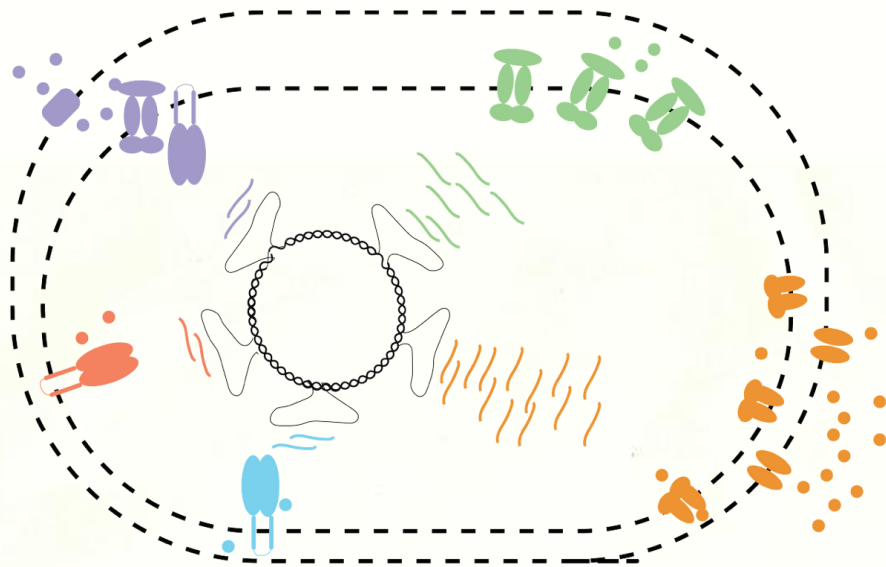


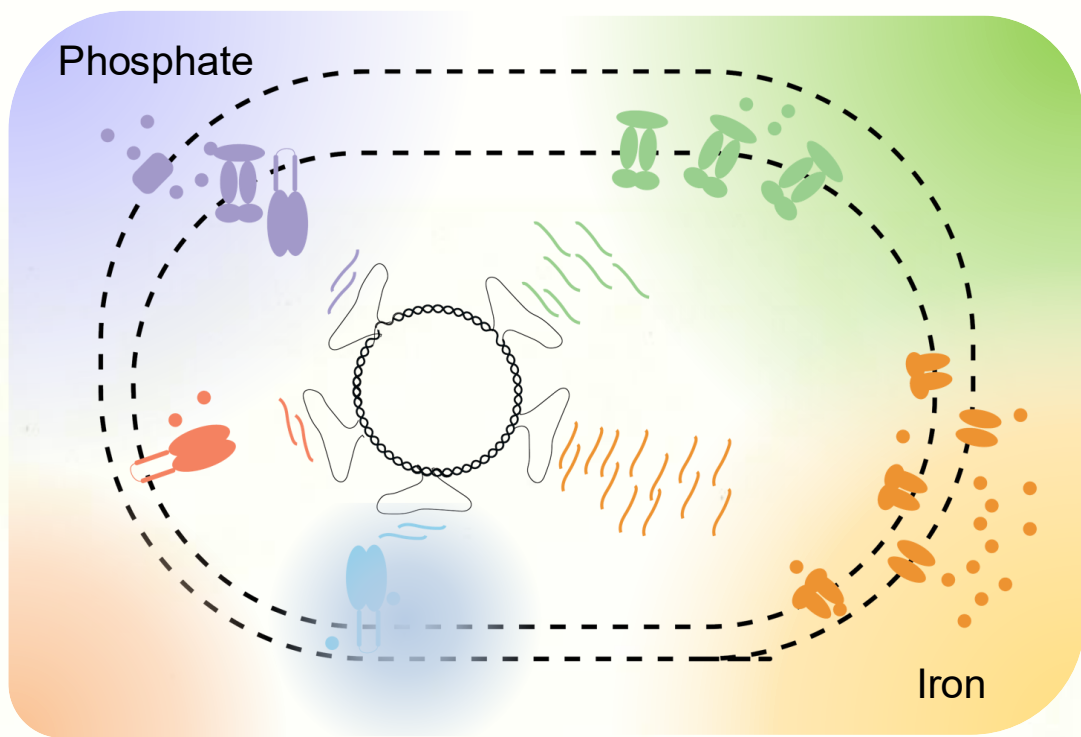




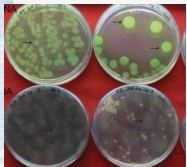
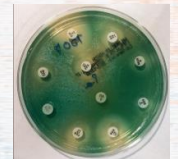
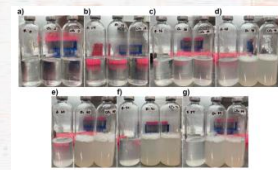
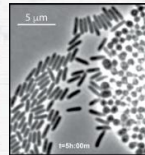
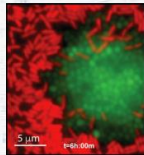
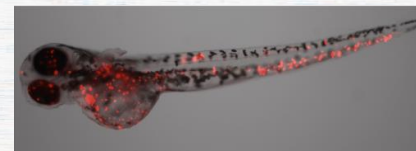
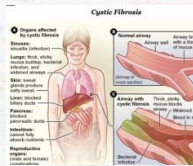
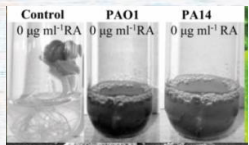


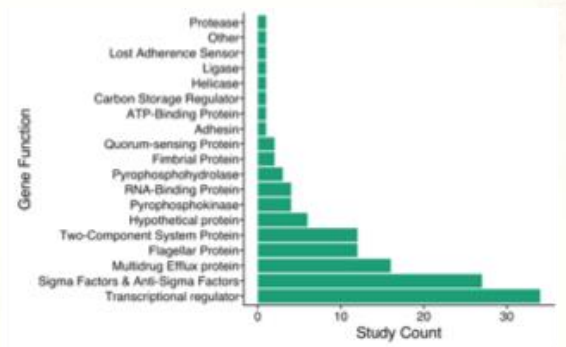


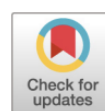












# Computationally Efficient Assembly of *Pseudomonas aeruginosa* Gene Expression Compendia

 Georgia Doing,<sup>a</sup>  Alexandra J. Lee,<sup>b</sup> Samuel L. Neff,<sup>a</sup> Taylor Reiter,<sup>c</sup> Jacob D. Holt,<sup>a</sup> Bruce A. Stanton,<sup>a</sup>  Casey S. Greene,<sup>c,d</sup>  
 Deborah A. Hogan<sup>a</sup>

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<sup>b</sup>Genomics and Computational Biology Graduate Program, University of Pennsylvania, Philadelphia, Pennsylvania, USA

<sup>c</sup>Department of Biochemistry and Molecular Genetics, University of Colorado School of Medicine, Denver, Colorado, USA

<sup>d</sup>Department of Systems Pharmacology and Translational Therapeutics, Perelman School of Medicine, University of Pennsylvania, Philadelphia, Pennsylvania, USA

<https://pubmed.ncbi.nlm.nih.gov/36541761/>

# ANTIMICROBIAL RESISTANCE THREATS

## in the United States, 2021-2022

CDC used new data<sup>1</sup> to analyze the U.S. burden of the following antimicrobial-resistant pathogens typically found in healthcare settings:



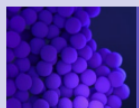
Carbapenem-resistant  
Enterobacteriales (CRE)



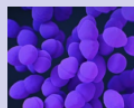
Carbapenem-resistant  
*Acinetobacter*



*Candida auris* (*C. auris*)



Methicillin-resistant  
*Staphylococcus aureus*  
(MRSA)



Vancomycin-resistant  
Enterococcus (VRE)



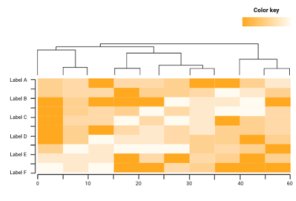
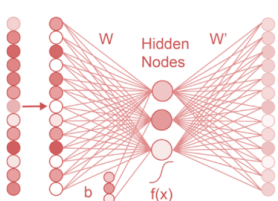
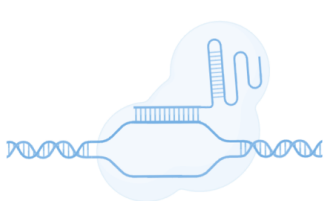
Extended-spectrum  
beta-lactamase (ESBL)-  
producing Enterobacteriales



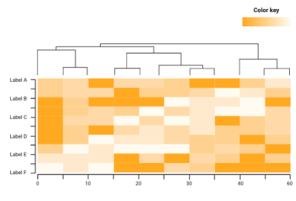
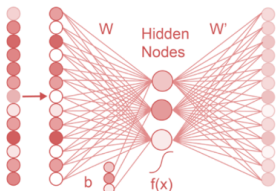
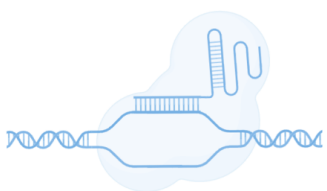
Multidrug-resistant (MDR)  
*Pseudomonas aeruginosa*

CDC previously reported that the burden of these pathogens increased in the United States in 2020 in the [COVID-19 Impact Report](#). The information below describes the burden in the two following years, 2021 and 2022, and compares against 2019 data.

CDC <https://www.cdc.gov/ecoli/about/index.html>

| SRA Search                                                                                                                                                                            | Compendia                                                                                                                                                  | DAE Models                                                                                                                                                                            | Bi-partite Graphs                                                                                                                                                                                                                                                                               | Comparative Genetics                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                      |                                                                           |                                                                                                      |                                                                                                                                                                                                               |                                                                                    |
| <ul style="list-style-type: none"> <li>organism: <i>E. coli</i></li> <li>molecule: mRNA</li> <li>pull data from Short Read Archive (SRA) and Gene Expression Omnibus (GEO)</li> </ul> | <ul style="list-style-type: none"> <li>metadata</li> <li><i>E. coli</i> (20 strains) reference pan-genomes</li> <li>salmon k-mer mapping (k=15)</li> </ul> | <ul style="list-style-type: none"> <li>denoising autoencoder (tied)</li> <li>1 layer of 50 hidden nodes</li> <li>fully connected</li> <li>objective): reconstruction error</li> </ul> | <ul style="list-style-type: none"> <li><math>G(E,V)</math> where <math>E \in \text{Corr}_{\text{gene, gene}}</math> and <math>V \in \text{Genes}_{\text{S.e.}} \cup \text{Genes}_{\text{S.a.}}</math></li> <li>Weight edges by correlation strength</li> <li>Color genes by homology</li> </ul> | <ul style="list-style-type: none"> <li>knock-down genes of interest under function-relevant conditions</li> <li>look for fitness effects and/or phenotypes</li> </ul> |



| SRA Search                                                                                                                                                                                                                    | Compendia                                                                                                                                                                              | DAE Models                                                                        | Bi-partite Graphs                                                                                 | Comparative Genetics                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|                                                                                                                                              |                                                                                                       |  |                 |           |
| <ul style="list-style-type: none"> <li>organism:<br/><i>E. coli</i></li> <li>molecule:<br/>DNA, RNA or text</li> <li>pull data from<br/>Short Read<br/>Archive (SRA)<br/>and Gene<br/>Expression<br/>Omnibus (GEO)</li> </ul> | <ul style="list-style-type: none"> <li>metadata</li> <li>subset of strains<br/>and samples</li> <li>reference pan-<br/>genomes</li> <li>salmon k-mer<br/>mapping<br/>(k=15)</li> </ul> | <p>Dimensionality<br/>reduction:</p> <p>PCA<br/>ICA<br/>NMD<br/>DAE</p>           | <p>Pattern<br/>detection:</p> <p>correlation<br/>regression<br/>clustering<br/>classification</p> | <p>Hypothesis<br/>testing:</p> <p>CRISPR data<br/>phenotypes<br/>(future<br/>directions)</p> |

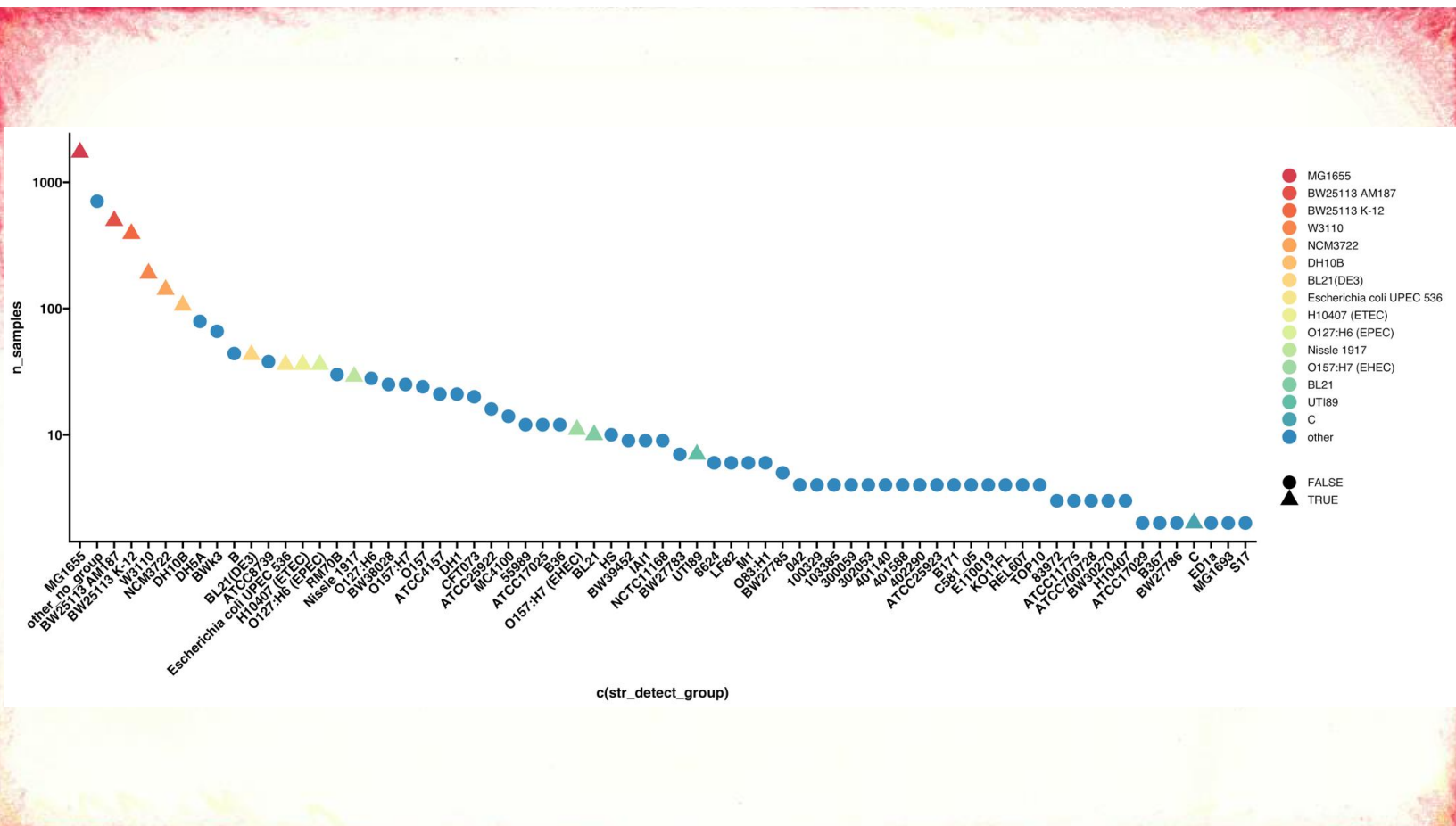
# Ecoli Pangenome Compendium

samples: 20,292

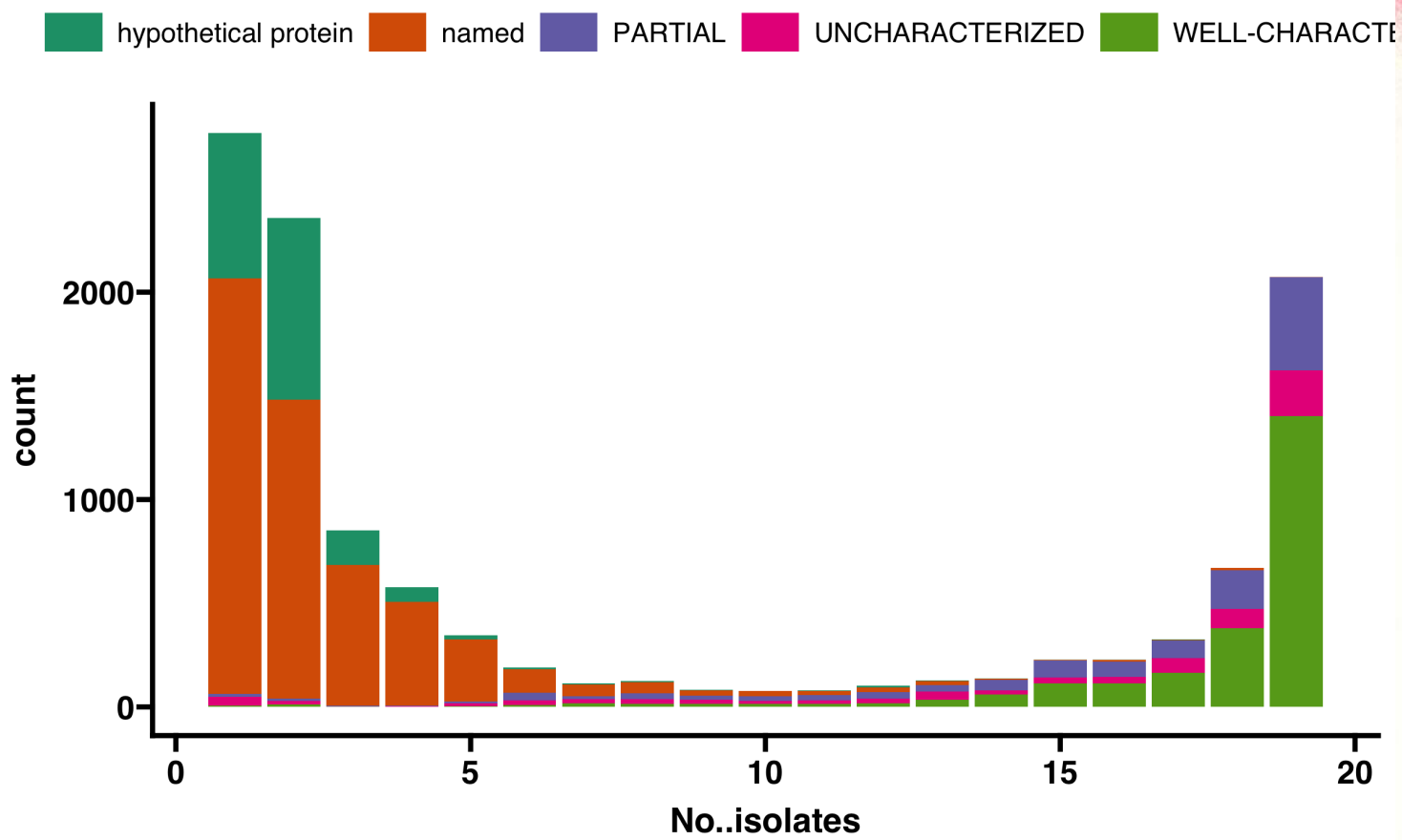
unique studies: 1,365

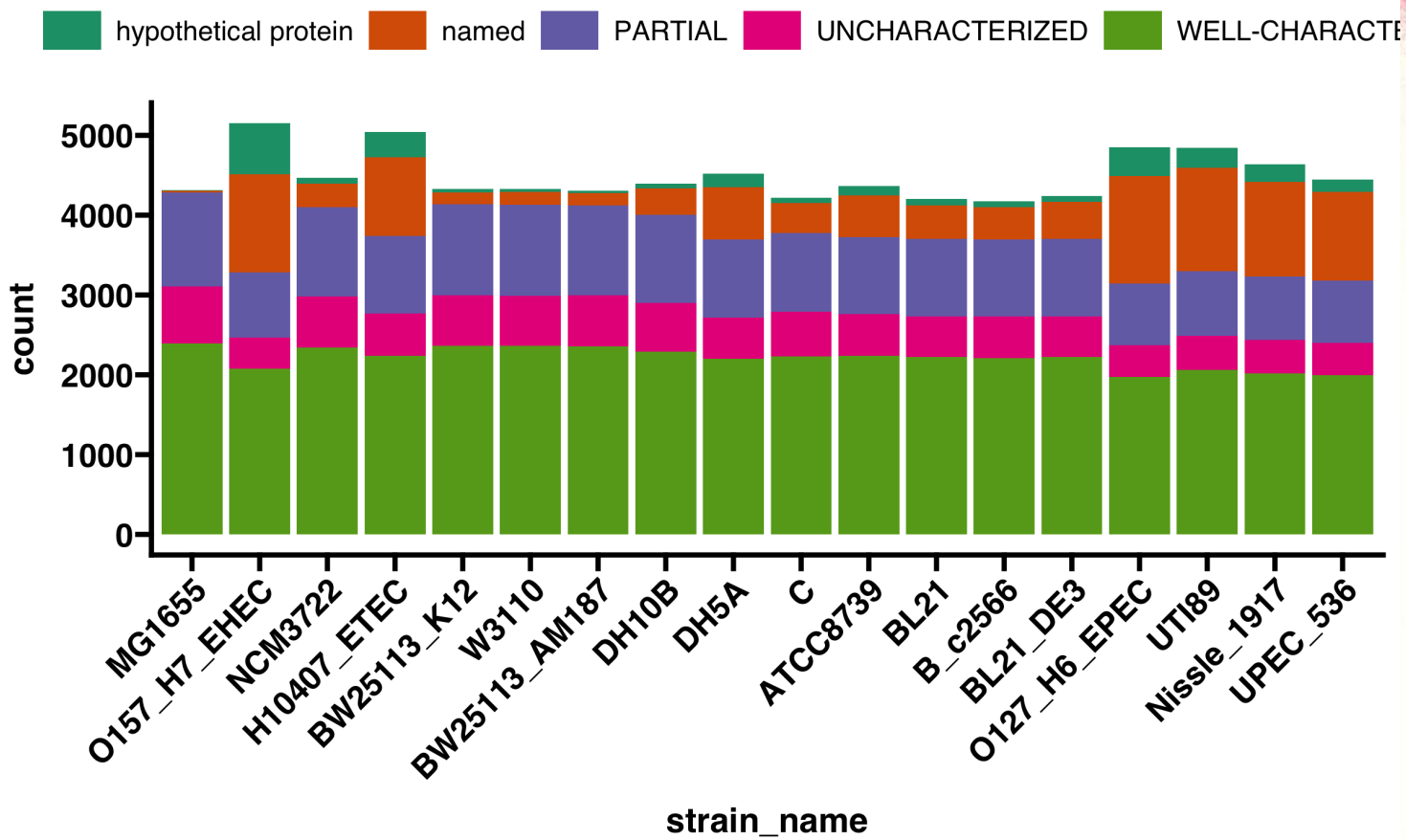
samples with high confidence strain: 4,260

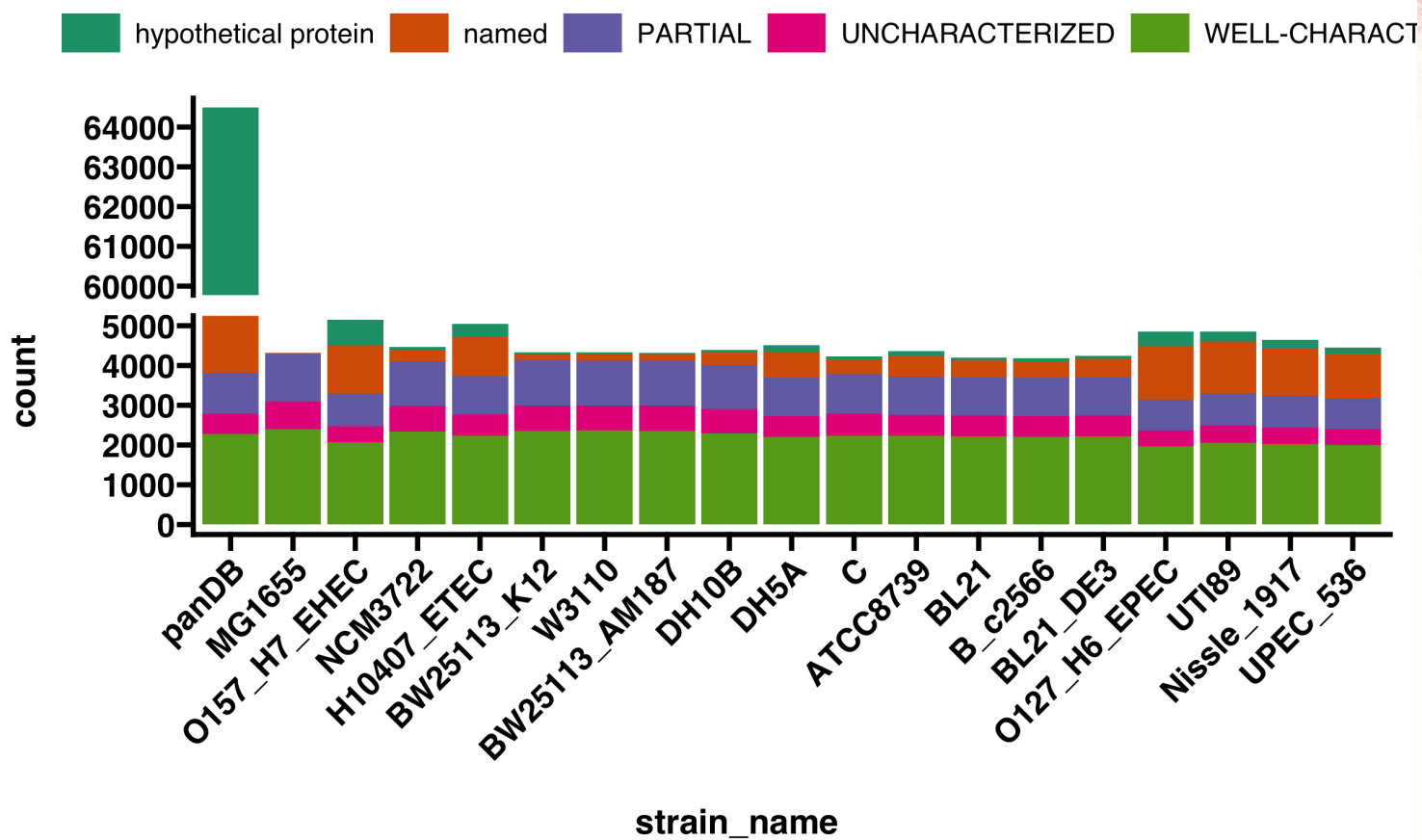
samples with putative strain: 11,859





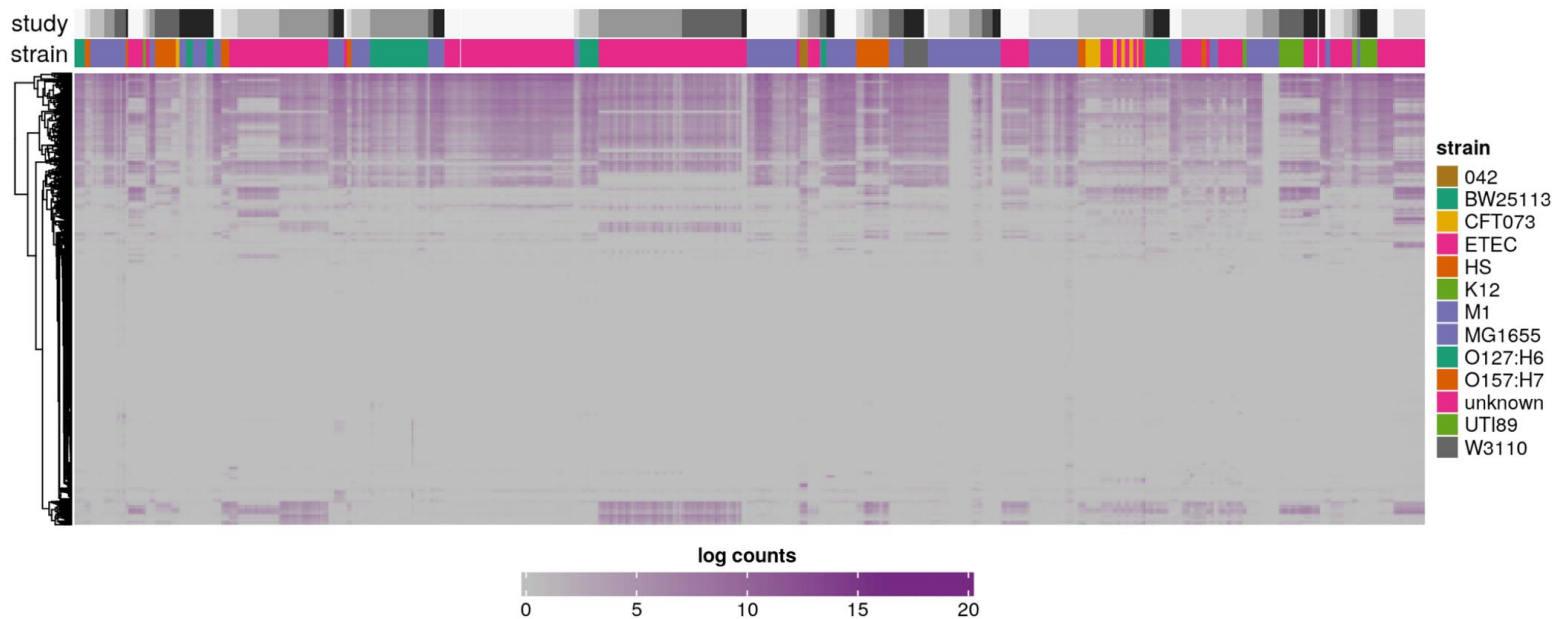


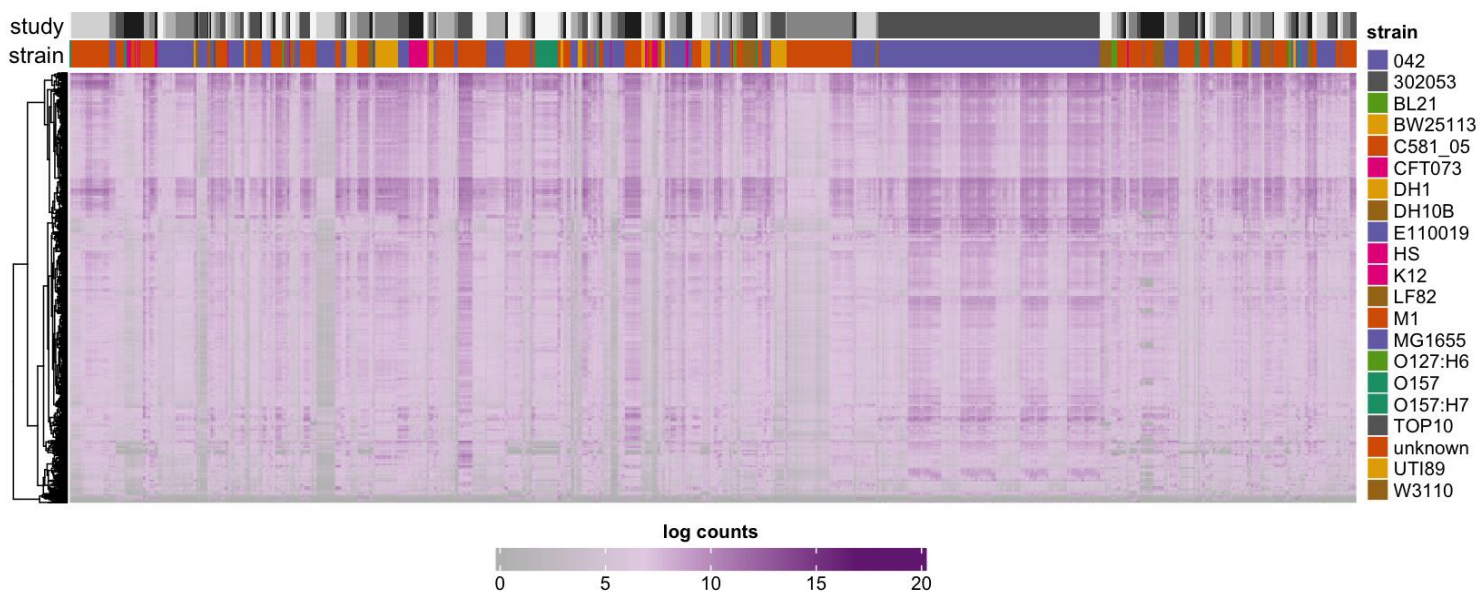




# Head of “count” table

| X                                     | Name         | ERX1805690 | ERX1805691 | ERX1805692 | ERX2279850 | ERX2279851 | ERX2279852 | ERX2279853 | ERX2279854 |
|---------------------------------------|--------------|------------|------------|------------|------------|------------|------------|------------|------------|
| 0ed86576d343ce011a61773e0620e335_1045 | character(0) | 1.000      | 0.000      | 0.00       | 0.000      | 0.000      | 0.000      | 0.000      | 0.000      |
| 0ed86576d343ce011a61773e0620e335_1047 | character(0) | 0.000      | 0.000      | 0.00       | 0.000      | 0.000      | 0.000      | 0.000      | 0.000      |
| 0ed86576d343ce011a61773e0620e335_1120 | character(0) | 0.000      | 0.000      | 0.00       | 6.476      | 6.236      | 0.000      | 0.000      | 7.554      |
| 0ed86576d343ce011a61773e0620e335_1175 | character(0) | 0.000      | 0.000      | 0.00       | 2.000      | 2.000      | 3.000      | 2.000      | 0.000      |
| 0ed86576d343ce011a61773e0620e335_1179 | character(0) | 0.000      | 0.000      | 0.00       | 2.000      | 4.000      | 0.000      | 4.000      | 2.000      |
| 0ed86576d343ce011a61773e0620e335_1311 | character(0) | 0.000      | 0.000      | 0.00       | 0.000      | 0.000      | 0.000      | 0.000      | 0.000      |
| 0ed86576d343ce011a61773e0620e335_1450 | character(0) | 0.000      | 0.000      | 0.00       | 3.000      | 0.000      | 0.000      | 0.000      | 2.000      |
| 0ed86576d343ce011a61773e0620e335_1476 | character(0) | 5.403      | 2.351      | 0.00       | 15.370     | 11.212     | 23.317     | 5.232      | 12.820     |
| 0ed86576d343ce011a61773e0620e335_1484 | character(0) | 0.000      | 9.229      | 10.55      | 2.000      | 2.000      | 0.000      | 0.000      | 3.000      |
| 0ed86576d343ce011a61773e0620e335_2071 | character(0) | 0.000      | 0.000      | 0.00       | 0.000      | 0.000      | 36.005     | 0.000      | 0.000      |





From DEseq2, median of ratios normalization:

$(Y_{ij})$ : The raw count for gene  $(i)$  in sample  $(j)$ .

$(g_i)$ : The geometric mean of counts for gene  $(i)$  across all samples.

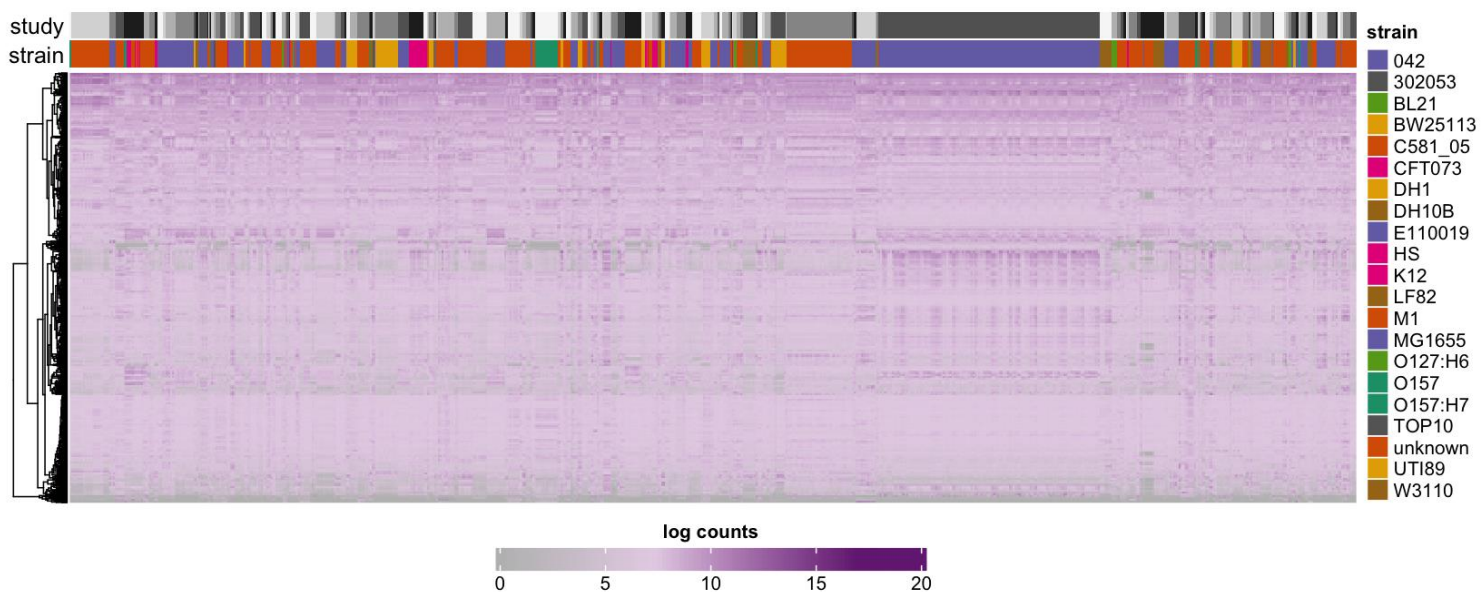
The size factor for sample  $(j)$ , denoted as  $(S_j)$ , is calculated as:

$$\left[ S_j = \text{median} \left( \frac{Y_{ij}}{g_i} \right)_{i \in 1, \dots, m} \right]$$

Where:

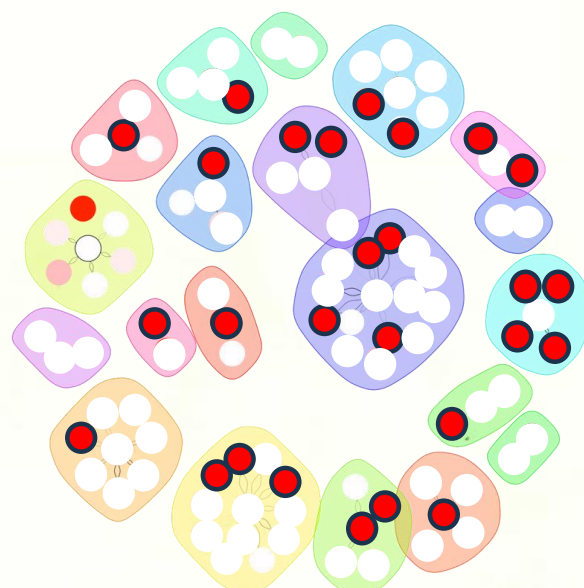
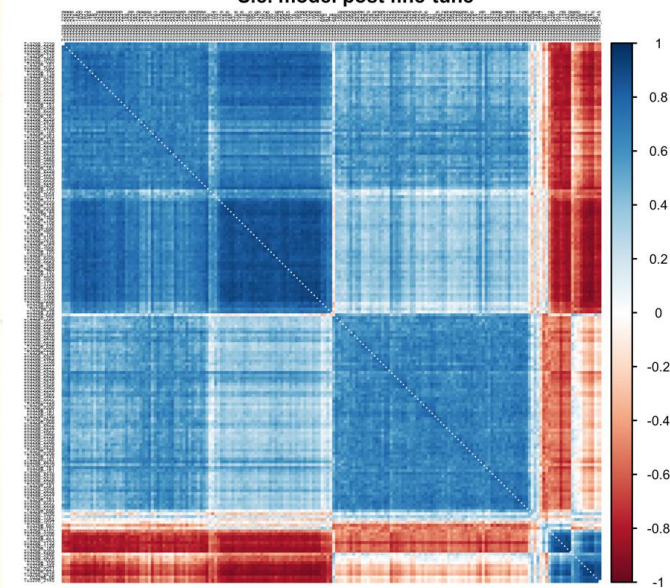
$(g_i = (\prod_{j=1}^n Y_{ij})^{1/n})$ , the geometric mean of counts for gene  $(i)$  across  $(n)$  samples.

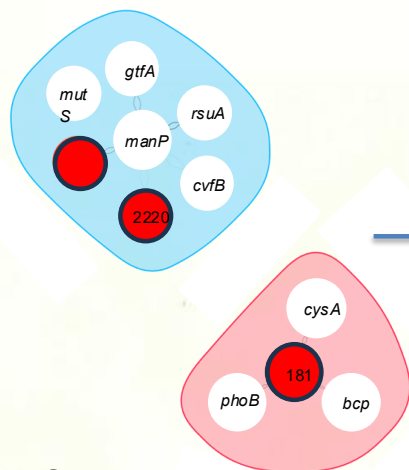
The ratio  $(\frac{Y_{ij}}{g_i})$  is computed for each gene  $(i)$  in sample  $(j)$ . The median of these ratios is taken as the size factor  $(S_j)$  for sample  $(j)$ .



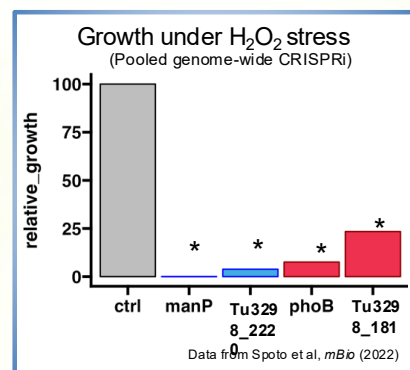
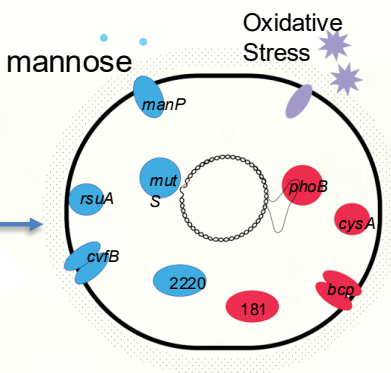


S.e. model post fine-tune





Uncharacterized gene  
 Gene with known function



## Lab machines: getting started

- Log on:
  - Your Union usernames
  - Default password: the word union followed by your ID number; e.g., union12345

**Always log off before leaving the lab**

## *Week 3*

### *Alignment link*

- alignment example:
- [https://bioboot.github.io/bimm143\\_W20/class-material/nw/](https://bioboot.github.io/bimm143_W20/class-material/nw/)

### *E coli metadata activity*

- colab notebook: [https://colab.research.google.com/drive/1X18-sF7Spht9ZXZlDkY2S\\_zvgEEf4xJG?usp=sharing](https://colab.research.google.com/drive/1X18-sF7Spht9ZXZlDkY2S_zvgEEf4xJG?usp=sharing)
- metadata 1: [https://drive.google.com/file/d/1BB7M3eWo8zPuxmlwSKMzE\\_RBeYrv1n0/view?usp=sharing](https://drive.google.com/file/d/1BB7M3eWo8zPuxmlwSKMzE_RBeYrv1n0/view?usp=sharing)
- metadata 2: [https://drive.google.com/file/d/18\\_PkR\\_sOLsQshSM4O1n2hVOcZoxgn8E0/view?usp=sharing](https://drive.google.com/file/d/18_PkR_sOLsQshSM4O1n2hVOcZoxgn8E0/view?usp=sharing)
- metadata 3: [https://drive.google.com/file/d/1\\_cJGV1gwHXzgUe7DMwHVI28nhJcma2tt/view?usp=sharing](https://drive.google.com/file/d/1_cJGV1gwHXzgUe7DMwHVI28nhJcma2tt/view?usp=sharing)

### *Link to data processing reading*

- original paper: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9948711/>
- review on gene expression compendia: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9396250/>

*Notes on reading scientific paper*

- from a previous Biological Data Science course at JAX
- colab notebook: [https://drive.google.com/file/d/1V7MIeRD7Lp8a2hEwKD\\_qLd67Vk2mSc24/view?usp=sharing](https://drive.google.com/file/d/1V7MIeRD7Lp8a2hEwKD_qLd67Vk2mSc24/view?usp=sharing)
- [Link to the slides](#)

*Khan Academy on the Central Dogma of Molecular Biology*

- from AP biology course
- <https://www.khanacademy.org/science/ap-biology/gene-expression-and-regulation/translation/a/intro-to-gene-expression-central-dogma>
- chapters on gene expression, the genetic code, translation and more

## *Week 4*

- link to gene sequence “[seqA](#)”
- [second notebook](#) introducing data
- [gene annotation data](#): presense/absence table for strains
- [raw transcription data](#): quasi-alignments counts from Salmon
- [column-wise normalized transcription data](#): TPM stands for ‘transcripts per million’, normalizes for different total read numbers per sample
- [scaled and filtered transcription data](#): filtered and normalized as in the paper, but for e coli



## *Week 5*

- Paper on dimensionality reduction: [Feldner-Busztin\\_et\\_al\\_2023](#)
- t-SNE visualizations: [Distill Blog](#)
- t-SNE on gene expression: [JEFworks github](#)
- continuing with the [second\\_data](#) introducing data.
- Dim reduction in single-cell RNAseq data: [Hackenberg et al 2025](#)
  
- Review on dimensionality reduction: [Wani 2025](#)
- Survey of dim reduction across scientific domains: [Cashman et al 2025](#)





## *Week 6*

### *Visualizations*

- still using [PANDAS](#)
- [matplotlib](#)
- [seaborn](#)
- [plotnine](#) package
- [plotnine\\_prism](#)
- [data-to-vis](#)
- [python graph gallery](#)
- the [second\\_data](#) notebook has been updated: <>
- I'll be continuing plotting examples in a fresh [plotting\\_data](#) notebook, updated to include Tuesday's notes.



## *Week 7*

- PyDESeq2 documentation: <https://pydeseq2.readthedocs.io/en/stable/>
- PYDeSeq2 paper: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10502239/>
- Roary Pangenome analysis (gene annotations and presence/absence data): <https://sanger-pathogens.github.io/Roary/>
- NCBI Run Tables (metadata): [Some Field Descriptions/Documentation](#)
- [hypothesis\\_testing](#) notebook from Tuesday
- “raw” data [https://drive.google.com/file/d/1IsrR-MFhrEWUFD45N5o\\_RcXVeQCYYyHV/view?usp=sharing](https://drive.google.com/file/d/1IsrR-MFhrEWUFD45N5o_RcXVeQCYYyHV/view?usp=sharing)



## *Week 8*

- Review on clustering: [Wani 2024](#)
- Conintuing in the [hypothesis\\_testing](#) notebook from week 7
- [Overleaf Template for ASM Journals](#)



## *Alignment Example*

*Needleman-Wunsch Alignment Example*



# CSC 483 - Needleman-Wunsch Example

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2026-01-21

## Intructions for the Needleman-Wunsch Alignment Algorithm

Given a mismatch scoring scheme  $s(x_i, y_j)$  where  $x_i$  and  $y_j$  are nucleotides (nt) and  $g$  is a gap penalty, the optimal alignmnet score of two nucleotide sequences  $X$  and  $Y$ , indexed by  $i$  and  $j$  respectively, is :

$$S(i, j) = \max \begin{cases} S(i-1, j-1) + s(x_i, y_j) & \text{(Match or Mismatch)} \\ S(i-1, j) + g & \text{(Gap in Y)} \\ S(i, j-1) + g & \text{(Gap in X)} \end{cases}$$

For example, setting  $g = -2$  and

$$s(x_i, y_j) = \begin{cases} 1 & \text{if Match} \\ -1 & \text{if Mismatch} \end{cases}$$

fill each square as:

$$S(i, j) = \max \begin{cases} \begin{cases} S(i-1, j-1) + 1 & \text{(Match)} \\ S(i-1, j-1) - 1 & \text{(Mismatch)} \end{cases} \\ S(i-1, j) - 2 & \text{(Gap in Y)} \\ S(i, j-1) - 2 & \text{(Gap in X)} \end{cases}$$

## Dynamic Programming Implementation

1. In the upper left corner, initiate with 0, then, move across columns and down rows. Fill in each square with:

- the maximum value given the values at the previous indices
- the previous indices from which the value builds (diagonal, left or up arrows)

|         | Seq $X$ |         | A       | G       | T           |
|---------|---------|---------|---------|---------|-------------|
| Seq $Y$ |         | $i = 0$ | $i = 1$ | $i = 2$ | $i = 3$     |
|         | $j = 0$ | 0       |         |         |             |
| A       | $j = 1$ |         |         |         |             |
| A       | $j = 2$ |         |         |         |             |
| G       | $j = 3$ |         |         |         |             |
| C       | $j = 4$ |         |         |         | final score |

2. Once all squares are filled, **the bottom right corner has your best possible score.**

### Example

**Step 1. Fill in scores for all  $i$  and  $j$**

For  $i = 1, j = 0$ :

$$S(i, j) = \max \begin{cases} S(0, j-1) + 1 = NA \text{ (indices cannot be } < 0) & \text{(Match)} \\ S(0, j-1) - 1 = NA \text{ (indices cannot be } < 0) & \text{(Mismatch)} \\ S(0, 0) - 2 = S(0, 0) - 2 = -2 & \text{(Gap in Y)} \\ S(1, j-1) - 2 = NA \text{ (indices cannot be } < 0) & \text{(Gap in X)} \end{cases}$$

$S(1, 0) = -2$ , from  $S(0, 0) = S(i-1, j)$ , so mark with a left arrow

|         | Seq $X$ |         | A       | G       | T       |
|---------|---------|---------|---------|---------|---------|
| Seq $Y$ |         | $i = 0$ | $i = 1$ | $i = 2$ | $i = 3$ |
|         | $j = 0$ | 0       | ← -2    |         |         |
| A       | $j = 1$ |         |         |         |         |
| A       | $j = 2$ |         |         |         |         |
| G       | $j = 3$ |         |         |         |         |
| C       | $j = 4$ |         |         |         |         |

Next, for  $i = 2, j = 0$ :

$$S(i, j) = \max \begin{cases} \begin{cases} S(1, j-1) + 1 = NA \text{ (indices cannot be } < 0) & \text{(Match)} \\ S(1, j-1) - 1 = NA \text{ (indices cannot be } < 0) & \text{(Mismatch)} \end{cases} \\ S(1, 0) - 2 = -2 - 2 = -4 & \text{(Gap in Y)} \\ S(2, j-1) - 2 = NA \text{ (indices cannot be } < 0) & \text{(Gap in X)} \end{cases}$$

$S(2, 0) = -4$ , from  $S(1, 0) = S(i-1, j)$ , so mark with a left arrow

|       | Seq X   |         | A       | G       | T       |
|-------|---------|---------|---------|---------|---------|
| Seq Y |         | $i = 0$ | $i = 1$ | $i = 2$ | $i = 3$ |
|       | $j = 0$ | 0       | ← -2    | ← -4    |         |
| A     | $j = 1$ |         |         |         |         |
| A     | $j = 2$ |         |         |         |         |
| G     | $j = 3$ |         |         |         |         |
| C     | $j = 4$ |         |         |         |         |

Note, that  $i = 3, j = 0$  follows similarly, as above. Give it a try, I've filled it in below.

Next, for  $i = 0, j = 1$ :

$$S(i, j) = \max \begin{cases} \begin{cases} S(i-1, 0) + 1 = NA \text{ (indices cannot be } < 0) & \text{(Match)} \\ S(i-1, 0) - 1 = NA \text{ (indices cannot be } < 0) & \text{(Mismatch)} \end{cases} \\ S(i-1, 1) - 2 = NA \text{ (indices cannot be } < 0) & \text{(Gap in Y)} \\ S(0, 0) - 2 = 0 - 2 = -2 & \text{(Gap in X)} \end{cases}$$

$S(0, 1) = -2$  derived from  $S(0, 0) = S(i, j-1)$  (up arrow)

|       | Seq X   |         | A       | G       | T       |
|-------|---------|---------|---------|---------|---------|
| Seq X |         | $i = 0$ | $i = 1$ | $i = 2$ | $i = 3$ |
|       | $j = 0$ | 0       | ← -2    | ← -4    | ← -6    |
| A     | $j = 1$ | ↑ -2    |         |         |         |
| A     | $j = 2$ |         |         |         |         |
| G     | $j = 3$ |         |         |         |         |
| C     | $j = 4$ |         |         |         |         |

The rest of the column ( $i = 0$ ) follows similarly.

To fill in the center squares, we start with  $i = 1, j = 1$ :

$$S(i, j) = \max \begin{cases} S(0, 0) + 1 = 0 + 1 = 1 & \text{(Match)} \\ S(0, 0) - 1 = NA \text{ (not a mismatch, } A=A) & \text{(Mismatch)} \\ S(0, 1) - 2 = -2 - 2 = -4 & \text{(Gap in } Y) \\ S(1, 0) - 2 = -2 - 2 = -4 & \text{(Gap in } X) \end{cases}$$

$S(1, 1) = 1$  from  $S(0, 0) = S(i - 1, j - 1)$ , so mark with a diagonal arrow

|          | Seq $X$ |               | <b>A</b>        | <b>G</b>        | <b>T</b>        |
|----------|---------|---------------|-----------------|-----------------|-----------------|
| Seq $Y$  |         | $i = 0$       | $i = 1$         | $i = 2$         | $i = 3$         |
|          | $j = 0$ | 0             | $\leftarrow -2$ | $\leftarrow -4$ | $\leftarrow -6$ |
| <b>A</b> | $j = 1$ | $\uparrow -2$ | $\nearrow 1$    |                 |                 |
| <b>A</b> | $j = 2$ | $\uparrow -4$ |                 |                 |                 |
| <b>G</b> | $j = 3$ | $\uparrow -6$ |                 |                 |                 |
| <b>C</b> | $j = 4$ | $\uparrow -8$ |                 |                 |                 |

Continuing, for  $i = 2, j = 1$ :

$$S(i, j) = \max \begin{cases} S(1, 0) + 1 = NA \text{ (not a match, } G \neq A) & \text{(Match)} \\ S(1, 0) - 1 = -2 - 1 = -3 & \text{(Mismatch)} \\ S(1, 1) - 2 = 1 - 2 = -1 & \text{(Gap in } Y) \\ S(2, 0) - 2 = -4 - 2 = -6 & \text{(Gap in } X) \end{cases}$$

$S(2, 1) = -1$  derived from  $S(1, 1) = S(i - 1, j)$  (left arrow)

|          | Seq $X$ |               | <b>A</b>        | <b>G</b>        | <b>T</b>        |
|----------|---------|---------------|-----------------|-----------------|-----------------|
| Seq $Y$  |         | $i = 0$       | $i = 1$         | $i = 2$         | $i = 3$         |
|          | $j = 0$ | 0             | $\leftarrow -2$ | $\leftarrow -4$ | $\leftarrow -6$ |
| <b>A</b> | $j = 1$ | $\uparrow -2$ | $\nearrow 1$    | $\leftarrow -1$ |                 |
| <b>A</b> | $j = 2$ | $\uparrow -4$ |                 |                 |                 |
| <b>G</b> | $j = 3$ | $\uparrow -6$ |                 |                 |                 |
| <b>C</b> | $j = 4$ | $\uparrow -8$ |                 |                 |                 |

Something to look out for, continuing, for  $i = 1, j = 2$ :

$$S(i, j) = \max \begin{cases} \begin{cases} S(0, 1) + 1 = -2 + 1 = -1 & \text{(Match)} \\ S(0, 1) - 1 = NA & \text{(Mismatch)} \end{cases} \\ S(0, 2) - 2 = -4 - 2 = -6 & \text{(Gap in Y)} \\ S(1, 1) - 2 = 1 - 2 = -1 & \text{(Gap in X)} \end{cases}$$

$S(1, 2) = -1$  could be from  $S(0, 1) = S(i - 1, j - 1)$  (diagonal arrow)...

OR....could be from  $S(1, 1) = S(i, j - 1)$  (up arrow).

|       | Seq X   |               | A                      | G               | T               |
|-------|---------|---------------|------------------------|-----------------|-----------------|
| Seq Y |         | $i = 0$       | $i = 1$                | $i = 2$         | $i = 3$         |
|       | $j = 0$ | 0             | $\leftarrow -2$        | $\leftarrow -4$ | $\leftarrow -6$ |
| A     | $j = 1$ | $\uparrow -2$ | $\nwarrow 1$           | $\leftarrow -1$ |                 |
| A     | $j = 2$ | $\uparrow -4$ | $\uparrow \nwarrow -1$ |                 |                 |
| G     | $j = 3$ | $\uparrow -6$ |                        |                 |                 |
| C     | $j = 4$ | $\uparrow -8$ |                        |                 |                 |

**Step 2. Find the final score in the bottom, right cell**

$$S(X, Y) = -1$$

|       | Seq X   |               | A                      | G               | T                        |
|-------|---------|---------------|------------------------|-----------------|--------------------------|
| Seq Y |         | $i = 0$       | $i = 1$                | $i = 2$         | $i = 3$                  |
|       | $j = 0$ | 0             | $\leftarrow -2$        | $\leftarrow -4$ | $\leftarrow -6$          |
| A     | $j = 1$ | $\uparrow -2$ | $\nwarrow 1$           | $\leftarrow -1$ | $\leftarrow -3$          |
| A     | $j = 2$ | $\uparrow -4$ | $\uparrow \nwarrow -1$ | $\nwarrow 0$    | $\leftarrow \nwarrow -2$ |
| G     | $j = 3$ | $\uparrow -6$ | $\uparrow -3$          | $\nwarrow 0$    | $\nwarrow -1$            |
| C     | $j = 4$ | $\uparrow -8$ | $\uparrow -5$          | $\uparrow -2$   | $\nwarrow -1$            |

**Step 3. Backtrace following the arrows for the optimal alignment**

- To figure out the alignment that leads to that best possible score, back-trace from the bottom right to the upper left, following the arrows of highest score attribution. Starting with the last nucleotide (3' end), pre-pend the nucleotides as you go.

- if diagonal arrow, record both nt at those indices (they may or may not match)

- if left arrow, record the nt from the top sequence (Seq  $X$ ,  $i$  indices) and a gap in the bottom sequence (Seq  $Y$ ,  $j$  indices)
- if up arrow, record the nt from the bottom sequence (Seq  $Y$ ,  $j$  indices) and a gap in the top sequence (Seq  $X$ ,  $i$  indices)

### Continuing the example

Starting in the bottom right cell  $i = 3, j = 4$ , given the diagonal arrow, we begin our backward alignment by recording both nucleotides at these indices:

```
Seq X:    T
          |
Seq Y:    C
```

This brings us to  $i = 2, j = 3$ , with another diagonal arrow so both nt are added again:

```
Seq X:    GT
          ||
Seq Y:    GC
Match:    *
```

Then to  $i = 1, j = 2$ , with a diagonal arrow **and** a left arrow, so we have two options:

Option 1, diagonal arrow, add both nt:

```
Seq X:    AGT
          |||
Seq Y:    AGC
Match:    **
```

Option 2, up arrow, add a gap to Seq2 ( $i$  indices):

```
Seq X:    -GT
          |||
Seq Y:    AGC
Match:    *
```

Finishing out the trace, we can choose an option (maybe follow the arrow that points to the higher score) or we can follow both options:

Option 1, now we are at  $i = 0, j = 1$  with an up arrow so add a gap to Seq2:

```
Seq X:    -AGT
          ||||
Seq Y:    AAGC
Match:    **
```

Option 2, now we are at  $i = 1, j = 1$  with a diagonal arrow so both nt get added:

```

Seq X:  A-GT
        ||||
Seq Y:  AAGC
Match:  *  *

```

Note that both alignments produces a score of -1 and both have 2 mismatches. If we broke this tie by following the path that always chooses the higher score, we would end with Option 2 since  $[S(1, 1) = 1] > [S(0, 1) = -2]$

You can see this using the online tool by setting your sequences to

```

Seq1 (Y): AAGC
Seq2 (X): AGT

```

[https://bioboot.github.io/bimm143\\_W20/class-material/nw/](https://bioboot.github.io/bimm143_W20/class-material/nw/)

If you "Clear Path" and manually trace Option 1, however, you will get the same score.

## Practice

Try this again with new sequences (any length), check your results using the online tool.

|       | Seq X   |         |         |         |         |         |         |         |         |         |
|-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Seq Y |         | $i = 0$ | $i = 1$ | $i = 2$ | $i = 3$ | $i = 4$ | $i = 5$ | $i = 6$ | $i = 7$ | $i = 8$ |
|       | $j = 0$ |         |         |         |         |         |         |         |         |         |
|       | $j = 1$ |         |         |         |         |         |         |         |         |         |
|       | $j = 2$ |         |         |         |         |         |         |         |         |         |
|       | $j = 3$ |         |         |         |         |         |         |         |         |         |
|       | $j = 4$ |         |         |         |         |         |         |         |         |         |
|       | $j = 5$ |         |         |         |         |         |         |         |         |         |
|       | $j = 6$ |         |         |         |         |         |         |         |         |         |
|       | $j = 7$ |         |         |         |         |         |         |         |         |         |
|       | $j = 8$ |         |         |         |         |         |         |         |         |         |

# *Reading Scientific Papers*

*Slides (html)*





about.qmd



## *References*

